# Multivariable Calculus

MATH 1920 Lecture Notes

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# 1 Vectors in Two and Three Dimensions

**Question:** What information specifies a line?

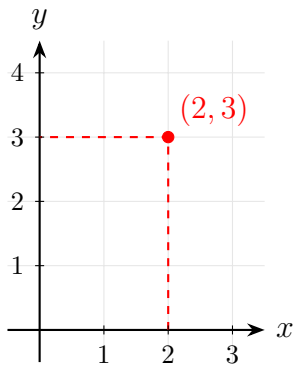
- 2 points
- A point and a direction vector
- Slope and $y$-intercept
- Intersection of 2 non-parallel planes

**Comparison:**

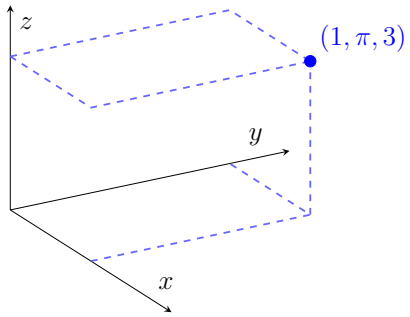
- **Single-variable calculus:** $f(x) \to$ rate of change of $x$
- **Multivariable calculus:** $f(x, y) \to$ functions of multiple variables

## 1.1 Coordinates

**2-dimension (2D):** $(2, 3)$




**3-dimension (3D):** $(1, \pi, 3)$




## 1.2 Vectors and Position Vectors

**Definition 1.1: Equivalent Vectors**

Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

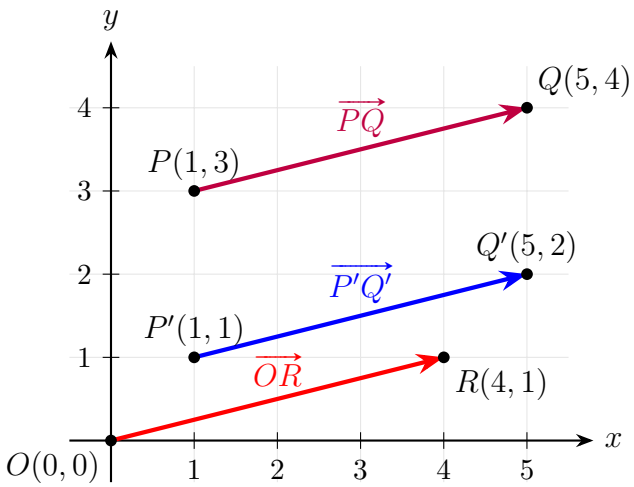
**Example 1.1.**

Given points $P = (1, 3)$, $Q = (5, 4)$ and $P' = (1, 1)$, $Q' = (5, 2)$. Let $R = (4, 1)$ and

$O = (0, 0)$. Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

This could be indicated by the diagram below:




**Definition 1.2: Position Vector**

A vector whose initial point is at the origin $O(0, 0)$ is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

**Definition 1.3: Magnitude**

The **length** or **magnitude** of a vector $\vec{v}$, denoted by $\|\vec{v}\|$, is the length of the line segment between its endpoints.

If $\vec{v} = \langle v_1, v_2 \rangle$, then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using $\overrightarrow{PQ} = \langle 4, 1 \rangle$:

$$\left\|\overrightarrow{PQ}\right\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

## 1.3 Unit Vectors

**Definition 1.4: Unit Vector**

A vector $\vec{v}$ is called a **unit vector** if its magnitude is 1, i.e., $\|\vec{v}\| = 1$.

In general, a unit vector of any function is given by:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|}$$

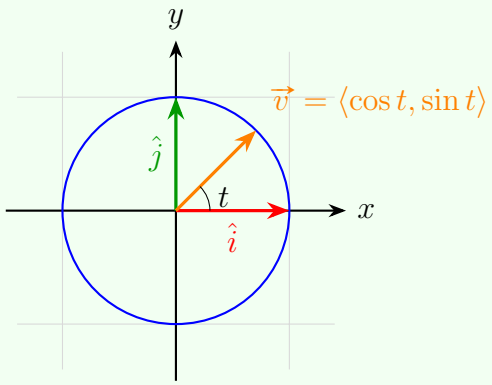
**Problem 1.1: Describing Unit Vectors**

Describe all unit vectors in $\mathbb{R}^2$.

**Solution 1.1**

A vector $\vec{v} = \langle x, y \rangle$ is a unit vector if and only if $x^2 + y^2 = 1$. Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t \rangle \quad \text{for } 0 \leq t < 2\pi$$




**Standard Basis Vectors:**

- **2D Basis Vectors:** $\hat{i} = \langle 1, 0 \rangle, \quad \hat{j} = \langle 0, 1 \rangle$
- **3D Basis Vectors:** $\hat{i} = \langle 1, 0, 0 \rangle, \quad \hat{j} = \langle 0, 1, 0 \rangle, \quad \hat{k} = \langle 0, 0, 1 \rangle$

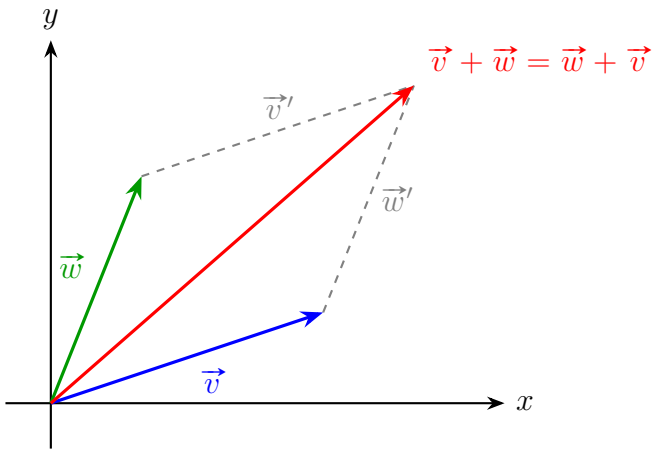
## 1.4 Vector Operations

**Addition and Subtraction:**

$$\langle v_1, v_2\rangle + \langle w_1, w_2\rangle = \langle v_1 + w_1, v_2 + w_2\rangle$$
$$\langle v_1, v_2\rangle - \langle w_1, w_2\rangle = \langle v_1 - w_1, v_2 - w_2\rangle \quad \text{(Equivalent to } \vec{v} + (-1)\vec{w}\text{)}$$

**Scalar Multiplication:** For any scalar $\lambda \in \mathbb{R}$ and vector $\vec{v} = \langle v_1, v_2\rangle$:

$$\lambda\vec{v} = \langle \lambda v_1, \lambda v_2\rangle$$




**Important Note**

Vectors are **not** designed to be divided! (Vector division is undefined).

## 1.5 Parallel Vectors

**Definition 1.5: Parallel Vectors**

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0\rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda\vec{v}$$

Equivalently, we have a geometric definition.

**Definition 1.6: Parallel Vectors (Geometric)**

$\vec{v} \parallel \vec{w}$ if and only if both vectors define or lie on the *same line passing through the origin.*

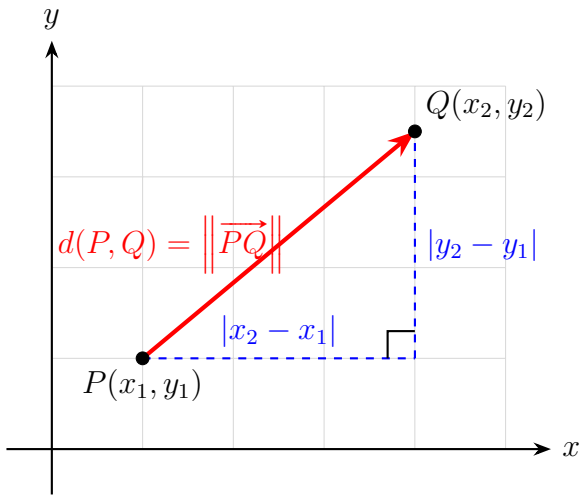
# 2 Discussion Week 1: Distance Formulae and Geometric Shapes

**Definition 2.1: Distance Formula in $\mathbb{R}^2$**

Let $(x, y) \in \mathbb{R}^2$ and points $P = (x_1, y_1)$, $Q = (x_2, y_2)$. The distance between $P$ and $Q$ is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance $d(P, Q)$ forms the hypotenuse of a right-angled triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$, as shown in the figure below.




The case in $\mathbb{R}^3$ follows similarly:

**Definition 2.2: Distance Formula in $\mathbb{R}^3$**

Let $(x, y, z) \in \mathbb{R}^3$ and points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2)$. The distance between $P$ and $Q$ is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

## 2.1 Equations of Geometric Shapes

Consider the equation $x^2 + y^2 = 4$.

In $\mathbb{R}^2$, this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius $r = 2$ centered at the origin. The distance from the origin $(0, 0)$ to any point $(x, y)$ on this circle is constant and equal to 2, which can be verified directly using the distance formula:

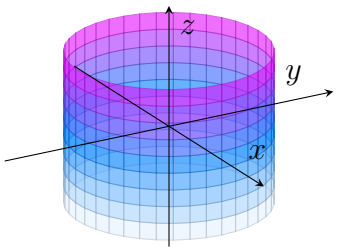
$$\sqrt{(x - 0)^2 + (y - 0)^2} = 2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation $x^2 + y^2 = 1$ in three-dimensional space, $\mathbb{R}^3$. This set is formally expressed as:

$$\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$$

Because $z$ is completely unrestricted, the cross-section at any constant height $z = c$ forms the unit circle $x^2 + y^2 = 1$. Consequently, this equation describes a *right circular cylinder* extending infinitely along the $z$-axis, as shown.




## 2.2 Linear Equations across Dimensions

In $\mathbb{R}^2$, the general form of a linear equation is given by:

$$Ax + By + C = 0 \quad (A, B, C \in \mathbb{R}, \text{ with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In $\mathbb{R}^3$, however, the exact same algebraic equation $Ax + By + C = 0$ represents a *plane* parallel to the $z$-axis (since $z$ is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).

# 3 Vectors and Directions

**Problem 3.1: Vector Directions**

Let $\vec{v} = \langle 3, -4, 2 \rangle$. Find a unit vector in the same direction as $\vec{v}$.

**Solution 3.1**

To find a unit vector in the same direction as $\vec{v}$, we first compute its magnitude:

$$\|\vec{v}\| = \sqrt{3^2 + (-4)^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

The unit vector in the same direction as $\vec{v}$ is then:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{29}} \langle 3, -4, 2 \rangle$$

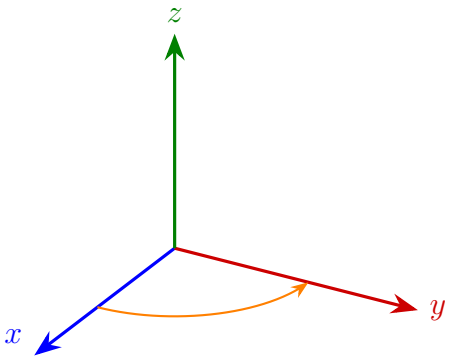
In general, for any non-zero vector $\vec{w}$,

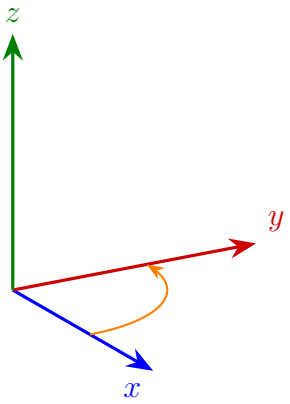
$$\begin{aligned} \|\lambda\vec{w}\| &= \sqrt{(\lambda w_1)^2 + (\lambda w_2)^2 + (\lambda w_3)^2} \\ &= \sqrt{\lambda^2(w_1^2 + w_2^2 + w_3^2)} \\ &= |\lambda|\|\vec{w}\| \end{aligned}$$

Hence, if we want a unit vector in the same direction as $\vec{w}$, we can choose $\lambda = \dfrac{1}{\|\vec{w}\|}$, giving us:

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$$

**Right-Hand Rule** The relative position of the $x$-, $y$-, and $z$-axes in $\mathbb{R}^3$ is determined by the *right-hand rule*. If you point your right thumb along the positive $x$-axis and your index finger along the positive $y$-axis, then your middle finger (perpendicular to both) will point along the positive $z$-axis.




**Definition 3.1: Parameterizations**

A parameterization is a function whose range is a given geometric object.

For example, a parameterization of the unit circle is given by:

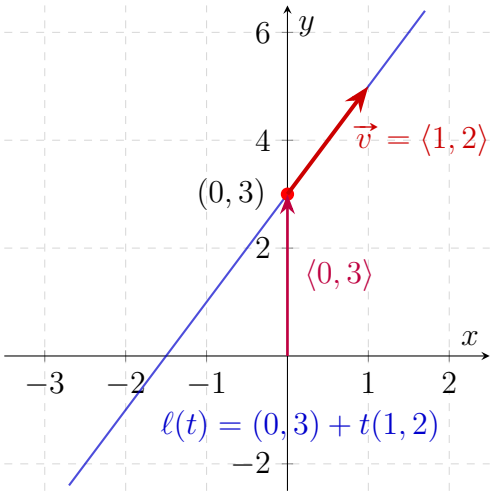
$$f(t) = (\cos t, \sin t)$$

**Example 3.1.**

To parameterize the line $y = 2x + 3$:

$$\begin{aligned} \ell(t) &= (t, 2t + 3) \\ &= \langle 0, 3 \rangle + \langle t, 2t \rangle \end{aligned}$$

The plot is then given:




To generalize, to parameterize a line in $\mathbb{R}^2$, we can first take a point on the line. Then, we

take some $\vec{v}$ in the direction of the line. Then, we can write the parameterization as:

$$\ell(t) = (p_1 + v_1 t,\ p_2 + v_2 t)$$

In $\mathbb{R}^3$, we can do the same thing.

**Example 3.2.**

Parameterize the line passing through $P(1, 2, 0)$ and $Q(5, -1, 1)$.

$$\begin{aligned}\overrightarrow{PQ} &= \langle 5-1, -1-2, 1-0\rangle \\ &= \langle 4, -3, 1\rangle \\ \ell(t) &= (1, 2, 0) + t(4, -3, 1) \\ &= (1 + 4t, 2 - 3t, t)\end{aligned}$$

## 3.1 Dot Product

**Definition 3.2: Dot Product**

The dot product of two vectors $\vec{v} = \langle v_1, v_2\rangle$ and $\vec{w} = \langle w_1, w_2\rangle$ is defined as:

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$

**Example 3.3.**

$$\vec{v} \cdot \vec{v} = v_1^2 + v_2^2 = \|\vec{v}\|^2$$
$$\vec{i} \cdot \vec{j} = 1 \cdot 0 + 0 \cdot 1 = 0$$

**Problem 3.2: Dot Product and Angle**

Find pairs of vectors with:

- Dot product $> 0$
- Dot product $= 0$

- Dot product $< 0$

**Solution 3.2**

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$ for any non-zero vector $\vec{v}$.
- $\vec{i} \cdot \vec{j} = 0$ since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$.

To conclude, the dot products of any two vectors corresponds to the following situations:

- Dot product $= 0$ if and only if the two vectors are orthogonal (perpendicular).
- Dot product $> 0$ if the angle between the two vectors is acute (less than 90 degrees).
- Dot product $< 0$ if the angle between the two vectors is obtuse (greater than 90 degrees).

This is because $\vec{v} \cdot \vec{w} = \|\vec{v}\|\|\vec{w}\| \cos\theta$, where $\theta$ is the angle between the vectors. Hence,

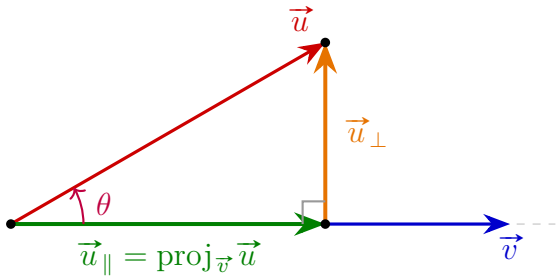
$$\theta = \arccos\left(\frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\|\|\vec{w}\|}\right)$$

**Definition 3.3: Orthogonal Projection**

Write $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$, where $\vec{u}_{\parallel} \parallel \vec{v}$ and $\vec{u}_{\perp} \perp \vec{u}_{\parallel}$. Then, $\vec{u}_{\parallel}$ is called the **orthogonal projection** of $\vec{u}$ onto $\vec{v}$.

**Example 3.4.**

Below is an illustration of orthogonal projection:



By this, we could see that

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

Then, we solve for $\vec{u}_{\parallel}$ and $\vec{u}_{\perp}$.

Taking the dot product with $\vec{v}$:

$$\begin{aligned}
\vec{u} \cdot \vec{v} &= (\vec{u}_{\parallel} + \vec{u}_{\perp}) \cdot \vec{v} \\
&= \vec{u}_{\parallel} \cdot \vec{v} + \vec{u}_{\perp} \cdot \vec{v} \\
&= \vec{u}_{\parallel} \cdot \vec{v} + 0 \quad (\text{since } \vec{u}_{\perp} \perp \vec{v}) \\
&= \vec{u}_{\parallel} \cdot \vec{v}
\end{aligned}$$

Let $\vec{e}_v = \dfrac{1}{\|\vec{v}\|}\vec{v}$ be the unit vector in the direction of $\vec{v}$, and write $\vec{u}_{\parallel} = \lambda \vec{e}_v$. Then:

$$\begin{aligned}
\vec{u} \cdot \vec{v} &= (\lambda \vec{e}_v) \cdot (\|\vec{v}\| \vec{e}_v) \\
&= \lambda \|\vec{v}\| (\vec{e}_v \cdot \vec{e}_v) \\
&= \lambda \|\vec{v}\| \|\vec{e}_v\|^2 \\
&= \lambda \|\vec{v}\| \quad (\text{since } \|\vec{e}_v\| = 1)
\end{aligned}$$

Solving for $\lambda$:

$$\lambda = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|}$$

Thus, the parallel projection vector $\vec{u}_{\parallel}$ and the orthogonal component $\vec{u}_{\perp}$ are given by:

$$\begin{aligned}
\vec{u}_{\parallel} &= \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}\right)\vec{e}_{v} \\
&= \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\vec{v} \\
&= \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v} \\
\vec{u}_{\perp} &= \vec{u} - \vec{u}_{\parallel} \\
&= \vec{u} - \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v}
\end{aligned}$$

Furthermore, the magnitude of the parallel projection is:

$$\left\|\vec{u}_{\parallel}\right\| = \|\vec{u}\||\cos\theta|$$

**Problem 3.3: Vector Projection**

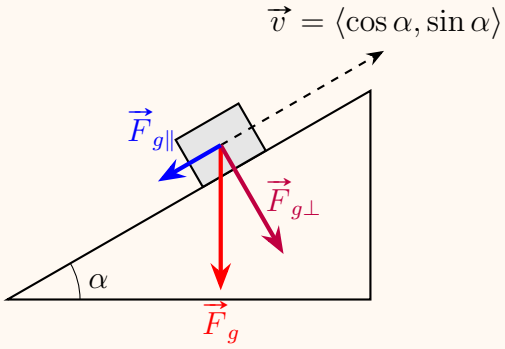
Find the projection of $\vec{u} = \langle 3, 2, 2\rangle$ onto $\vec{v}\,\langle 1, -2, 0\rangle$

**Solution 3.3**

$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{3\cdot 1 + 2\cdot(-1) + 2\cdot 0}{1^2 + (-1)^2 + 0^2}\langle 1, -1, 0\rangle \\
&= \frac{1}{2}\langle 1, -1, 0\rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

**Problem 3.4: Acceleration on an Inclined Plane**

Consider a block sliding down an inclined plane with an angle of inclination $\alpha$.




**Solution 3.4**

To find the magnitude of acceleration down the incline, we project the gravitational force $\vec{F}_g$ along the incline direction:

$$\begin{aligned}
\text{acceleration} &= \left\| \vec{F}_{g\parallel} \right\| \\
&= \left\| \vec{F}_g \right\| |\cos\theta| \\
&= \left\| \vec{F}_g \right\| \sin\alpha = 9.8 \sin\alpha \quad (\text{m/s}^2)
\end{aligned}$$

**Problem 3.5: Vector Projection**

Find the projection of $\vec{u} = \langle 3, 2, 2 \rangle$ onto $\vec{v} = \langle 1, -1, 0 \rangle$.

**Solution 3.5**

Using the vector projection formula:

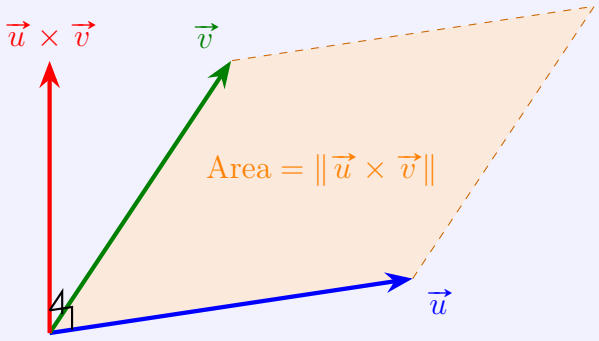
$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\
&= \frac{3(1) + 2(-1) + 2(0)}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\
&= \frac{3 - 2 + 0}{1 + 1 + 0} \langle 1, -1, 0 \rangle \\
&= \frac{1}{2} \langle 1, -1, 0 \rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

## 3.2 Determinants and Cross Products

**Definition 3.4: Vector Cross Product**

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$. The **cross product** $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$ satisfying the following three geometric properties:

1. **Orthogonality:** $\vec{u} \times \vec{v} \perp \vec{u}$ and $\vec{u} \times \vec{v} \perp \vec{v}$.
2. **Magnitude:** $\|\vec{u} \times \vec{v}\| = \|\vec{u}\|\|\vec{v}\| \sin\theta$, which equals the area of the parallelogram formed by $\vec{u}$ and $\vec{v}$.
3. **Orientation:** The ordered triple $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ is right-handed (positively oriented).




From Property (2), we have for the standard basis vectors:

$$\left\|\vec{i} \times \vec{i}\right\| = \left\|\vec{i}\right\|\left\|\vec{i}\right\| \sin 0^\circ = 0 \implies \vec{i} \times \vec{i} = \vec{0}$$

And by the Right-Hand Rule:

$$\vec{i} \times \vec{j} = \vec{k}$$

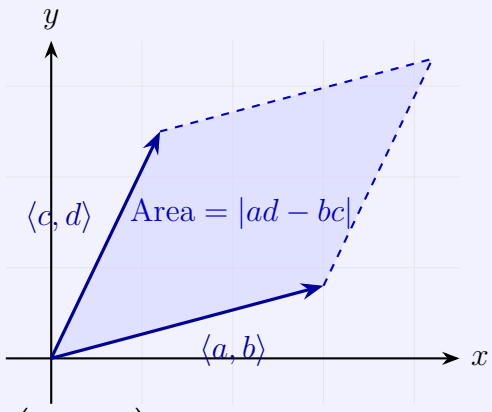
**Definition 3.5: Determinants**

For a $2 \times 2$ matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

This is denoted as $\det(A)$ or $|A|$ for a $2 \times 2$ matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

For a $2 \times 2$ matrix, the determinant represents the area of the parallelogram spanned by $\langle a, c \rangle$ and $\langle b, d \rangle$ (the column vectors) or $\langle a, b \rangle$ and $\langle c, d \rangle$.

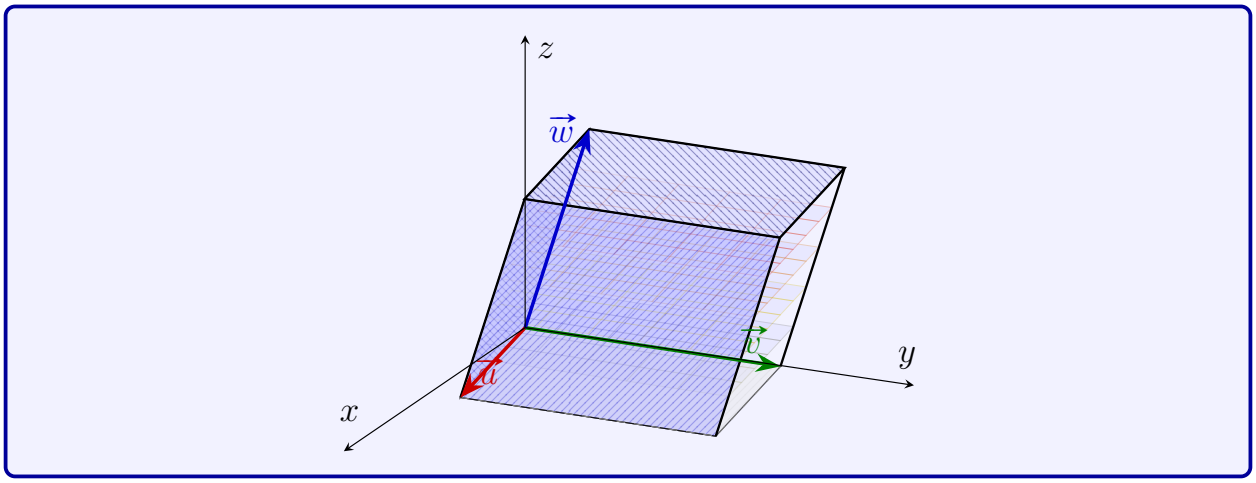



For a $3 \times 3$ matrix $B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, the determinant is defined as:

$$\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= aei + bfg + cdh - ceg - bdi - afh$$

$\det \begin{pmatrix} \vec{u} \\ \vec{v} \\ \vec{w} \end{pmatrix}$ represents the volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$.




The cross product of two vectors $\vec{u} = \langle u_1, u_2, u_3 \rangle$ and $\vec{v} = \langle v_1, v_2, v_3 \rangle$ can be computed using the determinant of a $3 \times 3$ matrix:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \vec{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \vec{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \vec{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

**Problem 3.6: Cross Product**

Let $\vec{v} = \langle 3, 2, -1 \rangle$, $\vec{w} = \langle 4, 1, -3 \rangle$. Find $\vec{v} \times \vec{w}$.

**Solution 3.6**

$$
\begin{aligned}
\vec{v} \times \vec{w} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 3 & 2 & -1 \\ 4 & 1 & -3 \end{vmatrix} \\
&= \vec{i} \begin{vmatrix} 2 & -1 \\ 1 & -3 \end{vmatrix} - \vec{j} \begin{vmatrix} 3 & -1 \\ 4 & -3 \end{vmatrix} + \vec{k} \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} \\
&= \vec{i}\,(2(-3) - (-1)(1)) - \vec{j}\,(3(-3) - (-1)(4)) + \vec{k}\,(3(1) - 2(4)) \\
&= \vec{i}\,(-6+1) - \vec{j}\,(-9+4) + \vec{k}\,(3-8) \\
&= \vec{i}\,(-5) - \vec{j}\,(-5) + \vec{k}\,(-5) \\
&= -5\vec{i} + 5\vec{j} - 5\vec{k}
\end{aligned}
$$

### 3.2.1 Properties of the Cross Product

- $\vec{v} \times \vec{w} \perp \vec{v}$ and $\vec{v} \times \vec{w} \perp \vec{w}$. This implies that $\vec{v} \cdot (\vec{v} \times \vec{w}) = 0$ and $\vec{w} \cdot (\vec{v} \times \vec{w}) = 0$.
- $\{\vec{v}, \vec{w}, \vec{v} \times \vec{w}\}$ forms a right-handed system.
- $\|\vec{v} \times \vec{w}\|$ represents the area of the parallelogram spanned by $\vec{v}$ and $\vec{w}$.

### 3.2.2 Algebraic Rules of the Cross Product

- Anti-commutativity: $\vec{v} \times \vec{w} = -(\vec{w} \times \vec{v})$
- Parallel vectors: $\vec{v} \times \vec{w} = \vec{0} \iff \vec{v} = \lambda\vec{w}$ or $\vec{w} = \vec{0}$.
- Scalar multiplication: $\lambda(\vec{v} \times \vec{w}) = (\lambda\vec{v}) \times \vec{w} = \vec{v} \times (\lambda\vec{w})$
- Distributivity: $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$
- Standard basis relations:

$$\vec{i} \times \vec{j} = \vec{k}, \quad \vec{j} \times \vec{k} = \vec{i}, \quad \vec{k} \times \vec{i} = \vec{j}$$

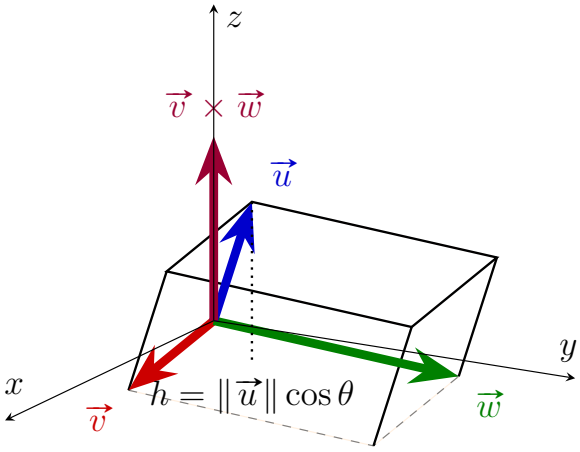
Hence, we may expand the cross product of two vectors $\vec{v} = \langle v_1, v_2, v_3 \rangle$ and $\vec{w} = \langle w_1, w_2, w_3 \rangle$ component-wise:

$$\begin{aligned}
\vec{v} \times \vec{w} &= (v_1 \vec{i} + v_2 \vec{j} + v_3 \vec{k}) \times (w_1 \vec{i} + w_2 \vec{j} + w_3 \vec{k}) \\
&= v_1 w_1 (\vec{i} \times \vec{i}) + v_1 w_2 (\vec{i} \times \vec{j}) + v_1 w_3 (\vec{i} \times \vec{k}) \\
&\quad + v_2 w_1 (\vec{j} \times \vec{i}) + v_2 w_2 (\vec{j} \times \vec{j}) + v_2 w_3 (\vec{j} \times \vec{k}) \\
&\quad + v_3 w_1 (\vec{k} \times \vec{i}) + v_3 w_2 (\vec{k} \times \vec{j}) + v_3 w_3 (\vec{k} \times \vec{k}) \\
&= v_1 w_2 \vec{k} - v_1 w_3 \vec{j} - v_2 w_1 \vec{k} + v_2 w_3 \vec{i} + v_3 w_1 \vec{j} - v_3 w_2 \vec{i} \\
&= (v_2 w_3 - v_3 w_2) \vec{i} + (v_3 w_1 - v_1 w_3) \vec{j} + (v_1 w_2 - v_2 w_1) \vec{k}
\end{aligned}$$

## 3.3 Scalar Triple Product

Given $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$, the quantity $\vec{u} \cdot (\vec{v} \times \vec{w})$ is defined as the **scalar triple product**.

The absolute value $|\vec{u} \cdot (\vec{v} \times \vec{w})|$ gives the geometric volume of the parallelepiped spanned by $\vec{u}$, $\vec{v}$, and $\vec{w}$:




The sign of $\vec{u} \cdot (\vec{v} \times \vec{w})$ indicates the orientation of the vector basis.

$\vec{u} \cdot (\vec{v} \times \vec{w}) > 0$ if the ordered triple $(\vec{u}, \vec{v}, \vec{w})$ forms a **right-handed system** (i.e., $\vec{u}$ points to the same side of the plane spanned by $\vec{v}$ and $\vec{w}$ as $\vec{v} \times \vec{w}$).

# 4 Discussion Week 2: Quadric Surfaces and Traces

**Definition 4.1: Quadric Surface**

A quadric surface is defined by a quadratic equation in three variables $x$, $y$, and $z$:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + ax + by + cz + d = 0$$

, where $A, B, C, D, E, F, a, b, c, d$ are constants, and at least one of $A, B, C, D, E, F$ is non-zero.

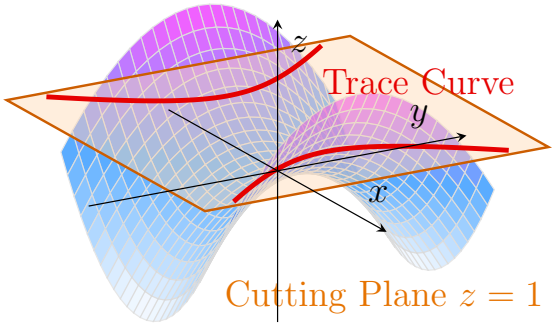
**Definition 4.2: Trace**

A trace is the intersection of a quadric surface with a plane, usually by taking slices parallel to the coordinate planes.

Note that when our surface can be written by $z = f(x, y)$, the traces form level sets of $f$.

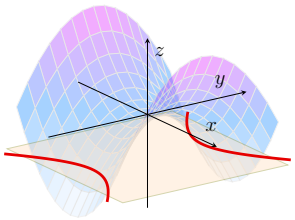
**Example 4.1.**

Consider the quadratic equation $z = x^2 - y^2$. Now take the traces in $z = -1$, $z = 0$, and $z = 1$. Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. In general, the traces in $x = k$ when $k$ is a constant are hypobolas with equations $z = k^2 - y^2$.

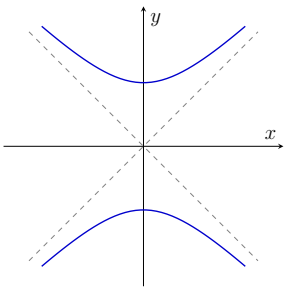



Now take the traces in $z = -1$, $z = 0$, and $z = 1$.

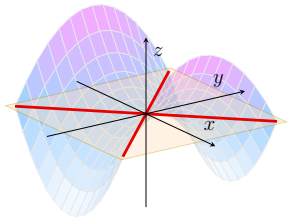
**Trace at $z = -1$**



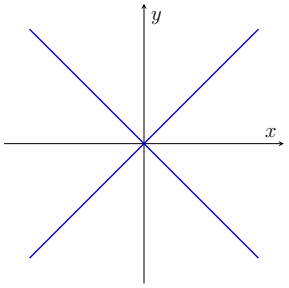

$y^2 - x^2 = 1$



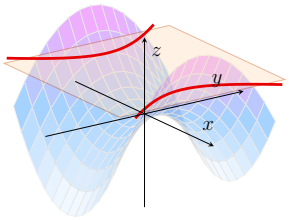

**Trace at $z = 0$**



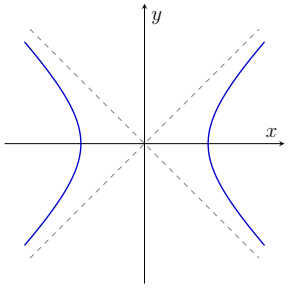

$y = \pm x$




**Trace at $z = 1$**

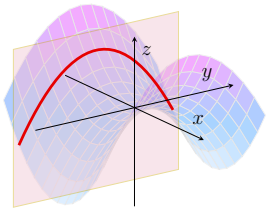



$x^2 - y^2 = 1$

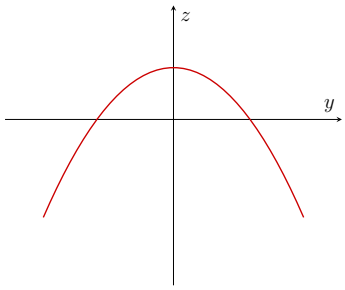



Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. Slicing the surface with vertical planes $x = k$ parallel to the $yz$-plane yields parabolic trace curves.

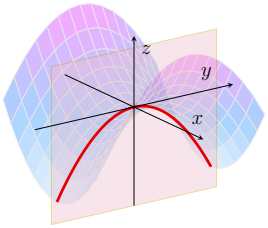
**Trace at $x = -1$**



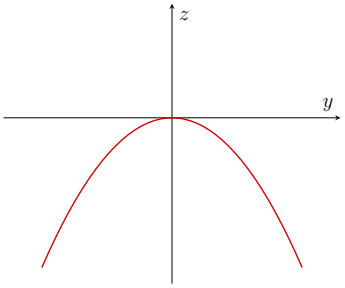

$z = 1 - y^2$



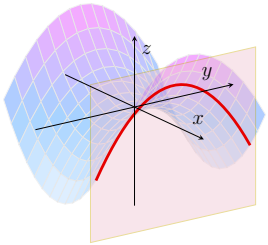

**Trace at $x = 0$**



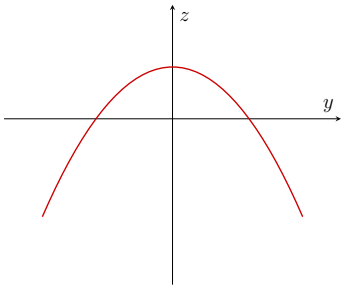

$z = -y^2$




**Trace at $x = 1$**




$z = 1 - y^2$




Ellipsoids are another family of quadric surfaces. The general equation of an ellipsoid is given by:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1$$

# 5 Planes and Surfaces

## 5.1 Determining a Plane

A plane $\mathcal{P}$ in $\mathbb{R}^3$ can be uniquely determined by any of the following geometric data:

- A point $Q$ on the plane and a normal vector $\vec{n} \perp \mathcal{P}$.
- Three non-collinear points.
- A general linear equation $ax + by + cz + d = 0$.
- Two intersecting lines (or two non-parallel direction vectors).
- Two parallel distinct lines.
- A line $\mathcal{L}$ and a point $Q \notin \mathcal{L}$.

## 5.2 Mathematical Formulations of a Plane

**Case 1: Plane passing through the origin** $O(0, 0, 0)$

A vector $\vec{v} = \langle x, y, z \rangle$ lies in $\mathcal{P} \iff \vec{v} \perp \vec{n}$, where $\vec{n} = \langle a, b, c \rangle$:

$$\vec{v} \in \mathcal{P} \iff \vec{v} \cdot \vec{n} = 0 \iff ax + by + cz = 0$$

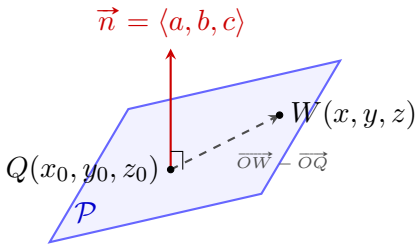
In set notation:

$$\mathcal{P} = \{\langle x, y, z \rangle \mid ax + by + cz = 0\}$$

**Case 2: Plane passing through a point $Q(x_0, y_0, z_0)$**

Let $Q$ be a known point on $\mathcal{P}$ with position vector $\overrightarrow{OQ} = \langle x_0, y_0, z_0 \rangle$. Any point $W(x, y, z)$ on the plane satisfies:

$$(\overrightarrow{OW} - \overrightarrow{OQ}) \perp \vec{n} \iff (\overrightarrow{OW} - \overrightarrow{OQ}) \cdot \vec{n} = 0$$




Expanding this dot product yields the point-normal form:

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

or equivalently:

$$ax + by + cz = ax_0 + by_0 + cz_0$$

We can always scale the coefficients $a, b, c, d$ in $ax + by + cz = d$ by $\frac{1}{\|\vec{n}\|}$ to obtain a unit normal vector $\hat{n} = \frac{\langle a,b,c \rangle}{\sqrt{a^2+b^2+c^2}}$.

## 5.3 Properties of Normal Vectors

For any set of vectors $\{\vec{u}, \vec{v}, \vec{w}\}$ where $\vec{v} \nparallel \vec{w}$, the following statements are equivalent:

- $\vec{u} \perp \vec{v}$ and $\vec{u} \perp \vec{w}$.
- $\vec{u} \parallel (\vec{v} \times \vec{w})$.
- $\vec{u} \perp \mathcal{P}$, where $\mathcal{P} = \text{Span}\{\vec{v}, \vec{w}\}$.

**Problem 5.1: Finding Normal Vectors**

Find a normal vector $\vec{n}$ to the plane $\mathcal{P}$ given by the equation $(x - 1) - (y - 1) = 0$.

**Solution 5.1**

The equation simplifies to $x - y = 0$. We can extract or compute the normal vector in two ways:

**Method 1 (Direct Inspection):** From the general form $ax + by + cz = d$, the coefficients of $x, y, z$ directly give the normal vector:

$$\vec{n} = \langle 1, -1, 0 \rangle$$

**Method 2 (Cross Product of Parallel Plane Vectors):** Two vectors parallel to the plane $\mathcal{P}$ are $\vec{v_1} = \langle 0, 0, 1 \rangle$ and $\vec{v_2} = \langle 1, 1, 0 \rangle$. Taking their cross product gives:

$$\vec{n} = \vec{v_1} \times \vec{v_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} = \vec{i}\,(0-1) - \vec{j}\,(0-1) + \vec{k}\,(0-0) = \langle -1, 1, 0 \rangle$$

Note that $\langle -1, 1, 0 \rangle = -\langle 1, -1, 0 \rangle$, which represents the same normal line direction.

**Problem 5.2: Finding a Plane From Three Points**

Find the equation of the plane containing the points $P(1, 0, 1)$, $Q(0, 2, 0)$, and $R(2, 1, 0)$.

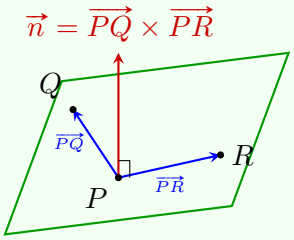
**Solution 5.2**

**Step 1: Construct two displacement vectors in the plane**

$$\overrightarrow{PQ} = \langle 0-1, 2-0, 0-1\rangle = \langle -1, 2, -1\rangle$$
$$\overrightarrow{PR} = \langle 2-1, 1-0, 0-1\rangle = \langle 1, 1, -1\rangle$$

**Step 2: Compute the normal vector via cross product**

$$\vec{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -1 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = \langle -1, -2, -3\rangle$$




**Step 3: Write the plane equation using point $P(1, 0, 1)$**

$$-1(x-1) - 2(y-0) - 3(z-1) = 0$$

Simplifying yields:

$$x + 2y + 3z = 4$$

# 6 Spheres and Planes

## 6.1 Determining a Sphere

A sphere in $\mathbb{R}^3$ can be uniquely determined by any of the following sets of data:

- A center point $(a, b, c) \in \mathbb{R}^3$ and a radius $R \in \mathbb{R}^{>0}$.
- Two points representing the endpoints of a diameter.
- Four non-coplanar points.

- A plane tangent to the sphere together with three points lying on the sphere.

## 6.2 Plane-Sphere Intersections

**Example 6.1.**

Consider the intersection of the sphere $x^2 + y^2 + z^2 = 4$ with the line $\ell(t) = (t, 0, 1)$ passing through $(1, 0, 1)$ and $(-1, 0, 1)$.

Substituting $\ell(t)$ into the sphere equation yields:

$$\begin{aligned} t^2 + 0^2 + 1^2 &= 4 \\ t^2 + 1 &= 4 \\ t^2 = 3 &\implies t = \pm\sqrt{3} \end{aligned}$$

Thus, the line intersects the sphere at $\ell(\sqrt{3}) = (\sqrt{3}, 0, 1)$ and $\ell(-\sqrt{3}) = (-\sqrt{3}, 0, 1)$.

## 6.3 Traces of Surfaces

**Definition 6.1: Traces**

A **trace** of a surface is the intersection curve obtained by intersecting the surface with a parallel coordinate plane of the form $x = k$, $y = k$, or $z = k$.

**Example 6.2** (Traces of a Sphere)**.**

Consider the sphere given by:

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

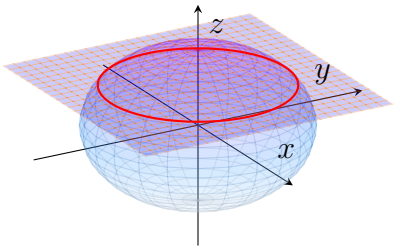
Taking the trace with the plane $z = k$ yields the equation:

$$(x - a)^2 + (y - b)^2 = R^2 - (k - c)^2$$

Depending on the plane distance $k$, we have three distinct cases:

- **Case 1 ($|k - c| = R$):** The trace is a single point at $(a, b, k)$.

- **Case 2 ($|k-c| < R$):** The trace is a circle of radius $\sqrt{R^2-(k-c)^2}$ centered at $(a, b, k)$.
- **Case 3 ($|k-c| > R$):** The trace is empty (no real intersection).




**Example 6.3.**

Find the trace of the plane $\mathcal{P} = \{(x, y, z) \mid -2x + 6y - 5z = 4\}$ with the plane $x = -1$:

$$\begin{aligned} -2(-1) + 6y - 5z &= 4 \\ 2 + 6y - 5z &= 4 \\ 6y - 5z = 2 \implies z &= \frac{6}{5}y - \frac{2}{5} \end{aligned}$$

The trace is a straight line in the plane $x = -1$.

## 6.4 Quadric Surfaces

A general **quadric surface** in $\mathbb{R}^3$ is given by the second-degree equation:

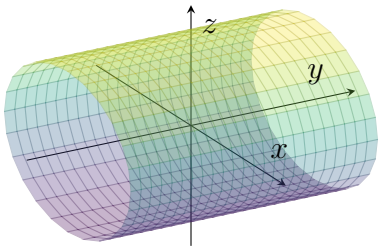
$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fzx + ax + by + cz + d = 0$$

When $D = E = F = 0$, the surface is in **standard position** (axes aligned with coordinate axes).

**Example 6.4.**

The surface $x^2 + z^2 = 1$ is a circular cylinder aligned along the $y$-axis with radius 1.

- $z = 0$ trace: $x^2 = 1 \implies x = \pm 1$ (parallel lines along $y$).
- $x = 0$ trace: $z^2 = 1 \implies z = \pm 1$ (parallel lines along $y$).

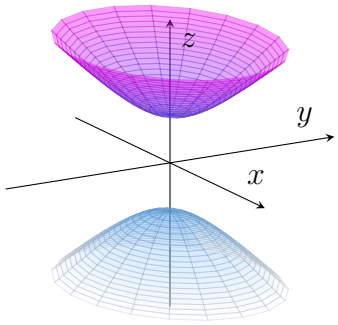



**Example 6.5.**

Consider the surface given by $x^2 + 4y^2 - z^2 = -1$.

**Traces:**

- $x = 0$ **trace:** $4y^2 - z^2 = -1 \implies z = \pm\sqrt{4y^2 + 1}$. As $|y| \to \infty$, $z \approx \pm 2y$ (Hyperbola).
- $y = 0$ **trace:** $x^2 - z^2 = -1 \implies z = \pm\sqrt{x^2 + 1}$. As $|x| \to \infty$, $z \approx \pm x$ (Hyperbola).
- $z = k$ **trace:** $x^2 + 4y^2 = k^2 - 1$:
  - $|k| = 1$: Single point at $(0, 0, \pm 1)$.
  - $|k| < 1$: No trace (empty set).
  - $|k| > 1$: Ellipse given by $x^2 + 4y^2 = k^2 - 1$.




**Why do cross-sectional ellipses fall on the surface?**

For any fixed height $z = k$ with $|k| > 1$, substituting $z = k$ fixes the value $k^2 - 1 > 0$ on the right side of $x^2 + 4y^2 = k^2 - 1$. This traces a locus of points $(x, y, k)$ whose horizontal distance components continuously satisfy the surface equation for that slice.

# 7 Vector-Valued Functions

## 7.1 Coordinate Systems in 3D Space

Specifying a point or giving directions in space can be accomplished in multiple ways:

- Giving a direction and a distance.
- Using Cartesian coordinates $(d_1, d_2, d_3) = (x, y, z)$.
- Specifying an angle relative to a reference axis along with a magnitude.
- Referencing a relative landmark.
- Using geographic coordinates like latitude and longitude.

### 7.1.1 Polar Coordinates (2D)

In the $xy$-plane, a point is specified by its radial distance $r$ from the origin and an angle $\theta$ relative to the positive $x$-axis: $(r, \theta)$.

$$r = \sqrt{x^2 + y^2} \quad (r \geq 0)$$

While $\theta = \arctan\left(\frac{y}{x}\right)$ is commonly used, it is undefined when $x = 0$. The foundational algebraic transformation between systems is given by:

$$x = r\cos\theta$$

$$y = r\sin\theta$$

### 7.1.2 Cylindrical Coordinates

Cylindrical coordinates $(r, \theta, z)$ extend polar coordinates into 3D space by preserving the standard Cartesian $z$-axis:

$$x = r\cos\theta$$
$$y = r\sin\theta$$
$$z = z$$

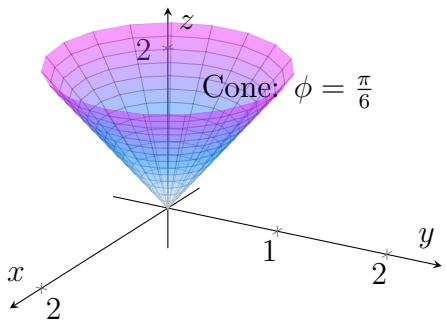
### 7.1.3 Spherical Coordinates

Spherical coordinates are represented as $(\rho, \theta, \phi)$, where $\rho$ is the distance from the origin, $\theta$ is the azimuthal angle in the $xy$-plane, and $\phi$ is the polar angle measured down from the positive $z$-axis.

$$x = \rho\sin\phi\cos\theta$$
$$y = \rho\sin\phi\sin\theta$$
$$z = \rho\cos\phi$$

The standard domain boundaries are:

$$\rho \geq 0$$
$$0 \leq \theta < 2\pi$$
$$0 \leq \phi \leq \pi$$




## 7.2 Coordinate System Transformations

**Problem 7.1: Conversion Between Coordinate Systems**

Convert the surface equation $x^2 + \dfrac{y^2}{a^2} = z^2$ into cylindrical and spherical coordinates.

**Solution 7.1**

**Cylindrical Coordinates:**

Substituting $x = r\cos\theta$, $y = r\sin\theta$, and $z = z$ into the original equation yields:

$$\begin{aligned}(r\cos\theta)^2 + \frac{(r\sin\theta)^2}{a^2} &= z^2 \\ r^2\cos^2\theta + \frac{r^2\sin^2\theta}{a^2} &= z^2 \\ r^2\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) &= z^2\end{aligned}$$

**Spherical Coordinates:**

Substituting $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$, and $z = \rho\cos\phi$ into the original equation yields:

$$\begin{aligned}(\rho\sin\phi\cos\theta)^2 + \frac{(\rho\sin\phi\sin\theta)^2}{a^2} &= (\rho\cos\phi)^2 \\ \rho^2\sin^2\phi\cos^2\theta + \frac{\rho^2\sin^2\phi\sin^2\theta}{a^2} &= \rho^2\cos^2\phi \\ \rho^2\sin^2\phi\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) &= \rho^2\cos^2\phi\end{aligned}$$

Assuming $\rho \neq 0$, we divide both sides by $\rho^2$:

$$\sin^2\phi\left(\cos^2\theta + \frac{\sin^2\theta}{a^2}\right) = \cos^2\phi$$
$$\cos^2\theta + \frac{\sin^2\theta}{a^2} = \frac{\cos^2\phi}{\sin^2\phi}$$
$$\cos^2\theta + \frac{\sin^2\theta}{a^2} = \cot^2\phi$$

## 7.3 Vector-Valued Functions

**Definition 7.1: Vector-Valued Functions**

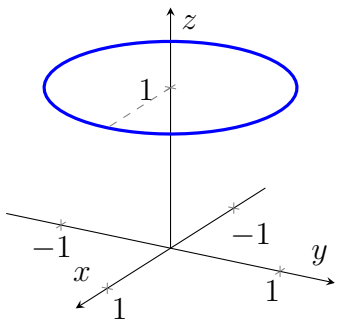
A vector-valued function maps one or more real parameters to a space vector:

$$\vec{r}(t) = \langle x(t), y(t), z(t)\rangle\,, \quad \vec{r} : I \to \mathbb{R}^3$$

where $I \subseteq \mathbb{R}$ is an interval. This traces out a spatial curve parametrized by $t$.

**Example 7.1.**

Consider the vector function $\vec{r}(t) = \langle \cos t, \sin t, 1\rangle$ for $0 \leq t < 2\pi$. This describes a unit circle in 3D space elevated to the horizontal plane $z = 1$.




**Problem 7.2: Finding Collisions and Intersections**

Let $\vec{r}_1(t) = \left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle$ and $\vec{r}_2(t) = \langle t, t^2\rangle$. Find all collision points and structural intersections between the curves.

**Solution 7.2**

**1. Collision Points (Same location at same time $t$):**

Setting $\vec{r}_1(t) = \vec{r}_2(t)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \langle t, t^2 \rangle$$

Equating the individual components gives the system:

$$t^2 + \frac{t}{2} = t$$
$$t - t^3 = t^2$$

Solving the first equation for $t$:

$$t^2 + \frac{t}{2} - t = 0$$
$$t^2 - \frac{t}{2} = 0$$
$$t\left(t - \frac{1}{2}\right) = 0 \implies t = 0 \quad \text{or} \quad t = \frac{1}{2}$$

Now, we verify both parameters in the second component equation $t - t^3 = t^2$:

$$\text{For } t = 0: \quad 0 - (0)^3 = (0)^2 \implies 0 = 0 \quad \text{(Valid)}$$
$$\text{For } t = \frac{1}{2}: \quad \frac{1}{2} - \left(\frac{1}{2}\right)^3 = \frac{1}{2} - \frac{1}{8} = \frac{3}{8}$$
$$\left(\frac{1}{2}\right)^2 = \frac{1}{4} = \frac{2}{8} \neq \frac{3}{8} \quad \text{(Invalid)}$$

Therefore, $t = 0$ is the only valid collision time. Evaluating $\vec{r}_1(0)$:

$$\vec{r}_1(0) = \left\langle 0^2 + \frac{0}{2}, 0 - 0^3 \right\rangle = \langle 0, 0 \rangle$$

The collision point is $(0, 0)$.

**2. Intersection Points (Same location at potentially different times $t$ and**

$s$**):**

Setting $\vec{r}_1(t) = \vec{r}_2(s)$ gives:

$$\left\langle t^2 + \frac{t}{2}, t - t^3 \right\rangle = \left\langle s, s^2 \right\rangle$$

Equating components yields:

$$s = t^2 + \frac{t}{2} \tag{1}$$

$$t - t^3 = s^2 \tag{2}$$

Substituting equation (1) into equation (2):

$$\begin{aligned} t - t^3 &= \left(t^2 + \frac{t}{2}\right)^2 \\ t - t^3 &= t^4 + t^3 + \frac{t^2}{4} \\ 0 &= t^4 + 2t^3 + \frac{t^2}{4} - t \\ 0 &= t\left(t^3 + 2t^2 + \frac{t}{4} - 1\right) \end{aligned}$$

Setting each factor to zero:

$$\begin{aligned} t = 0 &\implies s = 0^2 + \frac{0}{2} = 0 \\ &\implies \text{Intersection point } (0, 0) \\ t^3 + 2t^2 + \frac{t}{4} - 1 = 0 &\implies t \approx 0.576 \end{aligned}$$

Solving for $s$ when $t \approx 0.576$:

$$\begin{aligned} s &= (0.576)^2 + \frac{0.576}{2} \\ s &\approx 0.332 + 0.288 \approx 0.620 \end{aligned}$$

Evaluating the spatial coordinates using $\vec{r}_2(s)$:

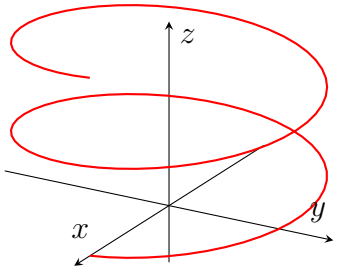
$$x = s \approx 0.620$$

$$y = s^2 \approx (0.620)^2 \approx 0.384$$

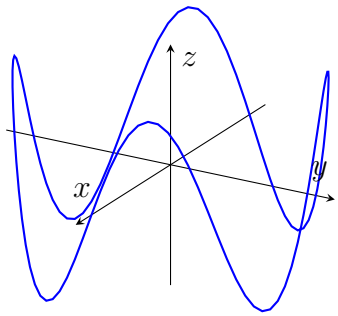
Hence, the two spatial intersection points are $(0, 0)$ and $(0.620, 0.384)$.

## 7.4 Gallery of Parametric Space Curves

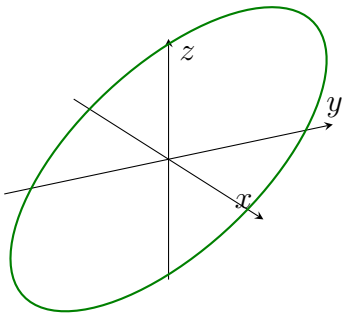
Below are standard parametric 3D space curves used in multivariable trajectory analysis:




$\vec{r}(t) = \langle \cos t, \sin t, t \rangle$




$\vec{r}(s) = \langle \cos s, \sin s, \sin(4s) \rangle$




$\vec{r}(t) = \langle \cos t, \sin t, 4 \sin t \rangle$

## 7.5 Parameterization of Curves and Surfaces

**Problem 7.3: Parameterization of a Curve on a Surface**

Find a parameterization for the curve given by the intersection of the surfaces $y = \sqrt{x^2 + z^2}$ and $x = z^2$.

**Solution 7.3**

Notice that for any choice of $z \in \mathbb{R}$, the value of $x$ is uniquely determined by $x = z^2$.

We can set $z = t$ as our parameter. Then,

$$x = t^2$$

$$z = t$$

$$y = \sqrt{(t^2)^2 + t^2} = \sqrt{t^4 + t^2} = |t|\sqrt{t^2 + 1}$$

Thus, a vector parameterization of the curve is:

$$\vec{r}(t) = \left\langle t^2, |t|\sqrt{t^2+1}, t \right\rangle, \quad t \in \mathbb{R}$$

## 7.6 Calculus of Vector-Valued Functions

Calculus operations on vector-valued functions are defined component-wise.

For a vector-valued function $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$, limits, derivatives, and integrals are computed as follows:

$$\lim_{t \to t_0} \vec{r}(t) = \left\langle \lim_{t \to t_0} x(t), \lim_{t \to t_0} y(t), \lim_{t \to t_0} z(t) \right\rangle$$

$$\vec{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$$

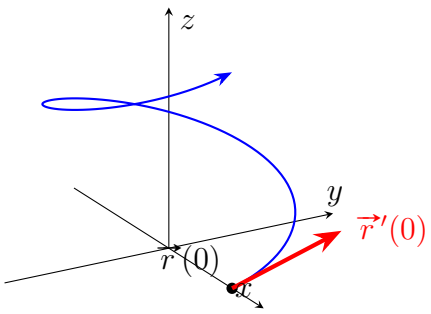
$$\int_a^b \vec{r}(t)\mathrm{d}t = \left\langle \int_a^b x(t)\mathrm{d}t, \int_a^b y(t)\mathrm{d}t, \int_a^b z(t)\mathrm{d}t \right\rangle$$

**Example 7.2** (Tangent Vectors and Parameterizing Tangent Lines)**.**
Given the circular helix $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$.

Differentiating component-wise gives:

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$$
$$\vec{r}'(0) = \langle 0, 1, 1 \rangle$$

The derivative $\vec{r}'(0)$ yields a vector representing the tangent direction of the curve at $t = 0$.




To parameterize the tangent line $\ell(t)$ to the helix at $t = 0$:

$$\vec{r}(0) = \langle 1, 0, 0 \rangle$$

Thus,

$$\begin{aligned} \ell(t) &= \vec{r}(0) + t\,\vec{r}'(0) \\ &= \langle 1, 0, 0 \rangle + t\,\langle 0, 1, 1 \rangle = \langle 1, t, t \rangle \end{aligned}$$

**Problem 7.4: Derivative and Singular Points of a Cycloid**

Find the derivative of the vector-valued function $\vec{r}(t) = \langle t - \sin t, 1 - \cos t \rangle$ and evaluate it at $t = \frac{\pi}{2}, \pi, 2\pi$.

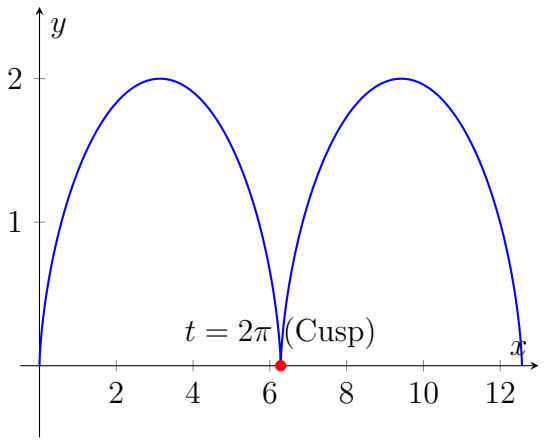
**Solution 7.4**

Differentiating each component with respect to $t$:

$$\begin{aligned} \vec{r}'(t) &= \left\langle \frac{\mathrm{d}}{\mathrm{d}t}(t - \sin t), \frac{\mathrm{d}}{\mathrm{d}t}(1 - \cos t) \right\rangle \\ &= \langle 1 - \cos t, \sin t \rangle \end{aligned}$$

Evaluating at specific parameter values:

$$\vec{r}'\left(\frac{\pi}{2}\right) = \left\langle 1 - \cos\frac{\pi}{2}, \sin\frac{\pi}{2} \right\rangle = \langle 1, 1 \rangle$$
$$\vec{r}'(\pi) = \langle 1 - \cos\pi, \sin\pi \rangle = \langle 2, 0 \rangle$$
$$\vec{r}'(2\pi) = \langle 1 - \cos 2\pi, \sin 2\pi \rangle = \langle 0, 0 \rangle$$




Notice from the graph of the cycloid that at $t = 2\pi$, the curve forms a sharp sharp corner (cusp). At this point, $\vec{r}'(2\pi) = \langle 0, 0 \rangle$. Although the vector function $\vec{r}(t)$ is differentiable everywhere, the tangent line direction is undefined at the cusp because the derivative vector vanishes.

## 7.7 Properties of Vector Derivatives & Integration

**Problem 7.5: Orthogonality Property for Constant-Length Vector Functions**

If $\|\vec{r}(t)\| = c$ for some constant $c$, what does this imply about the relationship between $\vec{r}(t)$ and its derivative $\vec{r}'(t)$?

**Solution 7.5**

Since $\|\vec{r}(t)\| = c$, taking the square of both sides yields:

$$\|\vec{r}(t)\|^2 = c^2 \implies \vec{r}(t) \cdot \vec{r}(t) = c^2$$

Differentiating both sides with respect to $t$ using the product rule for dot products:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\vec{r}(t)\cdot\vec{r}(t)\right)=\frac{\mathrm{d}}{\mathrm{d}t}(c^2)$$
$$\vec{r}\,'(t)\cdot\vec{r}(t)+\vec{r}(t)\cdot\vec{r}\,'(t)=0$$
$$2\vec{r}(t)\cdot\vec{r}\,'(t)=0 \implies \vec{r}(t)\cdot\vec{r}\,'(t)=0$$

Therefore, $\vec{r}(t)\perp\vec{r}\,'(t)$ for all $t$. (The position vector is always orthogonal to its velocity vector).

### Product Rule Verification for Dot Products

Given two vector-valued functions $\vec{r_1}(t),\vec{r_2}(t)\in\mathbb{R}^2$:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\left[\vec{r_1}(t)\cdot\vec{r_2}(t)\right]&=\frac{\mathrm{d}}{\mathrm{d}t}\left[x_1(t)x_2(t)+y_1(t)y_2(t)\right]\\&=x_1'(t)x_2(t)+x_1(t)x_2'(t)+y_1'(t)y_2(t)+y_1(t)y_2'(t)\\&=\vec{r_1}\,'(t)\cdot\vec{r_2}(t)+\vec{r_1}(t)\cdot\vec{r_2}\,'(t)\end{aligned}$$

**Problem 7.6: Initial Value Problem for Motion**

A particle moves with velocity $\vec{v}(t)=\left\langle 3t^2-2,\cos(\pi t),\sqrt{t}\right\rangle$. Find its position vector $\vec{r}(t)$, given the initial condition $\vec{r}(0)=\langle 2,1,1\rangle$.

**Solution 7.6**

Integrating $\vec{v}(t)$ component-wise:

$$\begin{aligned}\vec{r}(t)&=\int\vec{v}(t)\,\mathrm{d}t\\&=\left\langle\int(3t^2-2)\,\mathrm{d}t,\int\cos(\pi t)\,\mathrm{d}t,\int t^{1/2}\,\mathrm{d}t\right\rangle\\&=\left\langle t^3-2t+C_1,\frac{1}{\pi}\sin(\pi t)+C_2,\frac{2}{3}t^{3/2}+C_3\right\rangle\end{aligned}$$

Applying the initial condition $\vec{r}(0) = \langle 2, 1, 1\rangle$:

$$\vec{r}(0) = \langle C_1, C_2, C_3\rangle = \langle 2, 1, 1\rangle \implies C_1 = 2,\, C_2 = 1,\, C_3 = 1$$

Therefore,

$$\vec{r}(t) = \left\langle t^3 - 2t + 2, \frac{1}{\pi}\sin(\pi t) + 1, \frac{2}{3}t^{3/2} + 1 \right\rangle$$

*Note on Circular Motion:* If given acceleration $\vec{r}''(t) = \langle -\cos t, -\sin t, 0\rangle$, to uniquely guarantee that $\vec{r}(t)$ describes a circle centered at the origin, appropriate initial conditions for both position $\vec{r}(0)$ and velocity $\vec{r}'(0)$ must be specified.

## 7.8 Arc Length of Curves

The arc length $s$ of a smooth curve $\vec{r}(t) = \langle x(t), y(t)\rangle$ on $t \in [a, b]$ is derived as the limit of polygonal approximations:

$$s = \lim_{\Delta t \to 0} \sum \sqrt{\left(\frac{\Delta x}{\Delta t}\right)^2 + \left(\frac{\Delta y}{\Delta t}\right)^2}\,\Delta t = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\,\mathrm{d}t = \int_a^b \|\vec{r}'(t)\|\mathrm{d}t$$

**Example 7.3** (Arc Length of a Helix)**.**
Find the arc length of $\vec{r}(t) = \langle \cos t, \sin t, t\rangle$ for $t \in [0, 2\pi]$.

First, evaluate the speed $\|\vec{r}'(t)\|$:

$$\vec{r}'(t) = \langle -\sin t, \cos t, 1\rangle \implies \|\vec{r}'(t)\| = \sqrt{(-\sin t)^2 + \cos^2 t + 1^2} = \sqrt{2}$$

Thus, the arc length is:

$$s = \int_0^{2\pi} \sqrt{2}\,\mathrm{d}t = 2\pi\sqrt{2}$$

**Example 7.4** (Arc Length of an Ellipse and Elliptic Integrals)**.**
Consider the ellipse $x^2 + \frac{y^2}{4} = 1$.

**Method 1 (Cartesian Parameterization - Upper Half):** Setting $x = t$ and $y =$

$2\sqrt{1-t^2}$ for $t \in [-1, 1]$ yields:

$$\vec{r}(t) = \left\langle t, 2\sqrt{1-t^2} \right\rangle \implies \vec{r}\,'(t) = \left\langle 1, -\frac{2t}{\sqrt{1-t^2}} \right\rangle$$

$$\|\vec{r}\,'(t)\| = \sqrt{1 + \frac{4t^2}{1-t^2}} = \sqrt{\frac{1+3t^2}{1-t^2}}$$

The arc length of the upper half is given by

$$\int_{-1}^{1} \sqrt{\frac{1+3t^2}{1-t^2}}\, \mathrm{d}t.$$

**Method 2 (Trigonometric Parameterization - Full Ellipse):** Setting $x = \cos t$ and $y = 2\sin t$ for $t \in [0, 2\pi]$:

$$\vec{r}\,'(t) = \langle -\sin t, 2\cos t \rangle \implies \|\vec{r}\,'(t)\| = \sqrt{\sin^2 t + 4\cos^2 t} = \sqrt{4 - 3\sin^2 t}$$

The total perimeter of the ellipse is:

$$s = \int_0^{2\pi} \sqrt{4 - 3\sin^2 t}\, \mathrm{d}t$$

**Definition 7.2: Elliptic Integral**

An elliptic integral is a type of integral that cannot be expressed in terms of elementary functions.

*Elliptic Integrals:* It turns out these integrals cannot be evaluated in terms of elementary functions. They belong to a class of functions known as **elliptic integrals**, which require numerical methods to evaluate.

# 8 Scalar-Valued Functions of Several Variables

**Definition 8.1: Scalar Function**

A scalar-valued function of several variables maps a vector of multiple inputs to a single real output $f : D \subseteq \mathbb{R}^n \to \mathbb{R}$.

**Examples:**

- Temperature distribution: $T(\text{longitude}, \text{latitude}, \text{time}) : \mathbb{R}^3 \to \mathbb{R}$.
- Precipitation depth: $P(\text{longitude}, \text{latitude}) : \mathbb{R}^2 \to \mathbb{R}$.
- Domain and Range example: For $f(x, y) = y\sqrt{x-1}$,

$$\text{Domain: } \{(x, y) \in \mathbb{R}^2 : x \geq 1\}, \quad \text{Range: } \mathbb{R}$$

## 8.1 Visualizing Multivariable Functions: Traces and Level Sets

The graph of a function $f(x, y)$ is defined as the surface $\{(x, y, z) \in \mathbb{R}^3 : z = f(x, y)\}$.

**Example 8.1** (Analyzing Traces of $f(x, y) = y \sin x$)**.**

To visualize $f(x, y) = y \sin x$, we examine slices along constant coordinates (traces):

- **$y$-traces ($y = c$):**

  For $y = 0 \implies z = 0$;

  for $y = 1 \implies z = \sin x$;

  for $y = -1 \implies z = -\sin x$. (Sinusoidal curves scaled by $y$).

- **$x$-traces ($x = c$):**

  For $x = 0 \implies z = 0$;

  for $x = \frac{\pi}{2} \implies z = y$;

  for $x = \pi \implies z = 0$. (Lines whose slopes depend on $\sin x$).

## 8.2 Level Sets and Contour Maps

**Definition 8.2: Level Set**

For a function $f : D \to \mathbb{R}$ ($D \subseteq \mathbb{R}^2$), the **level set** (or level curve) at height $c$ is defined as

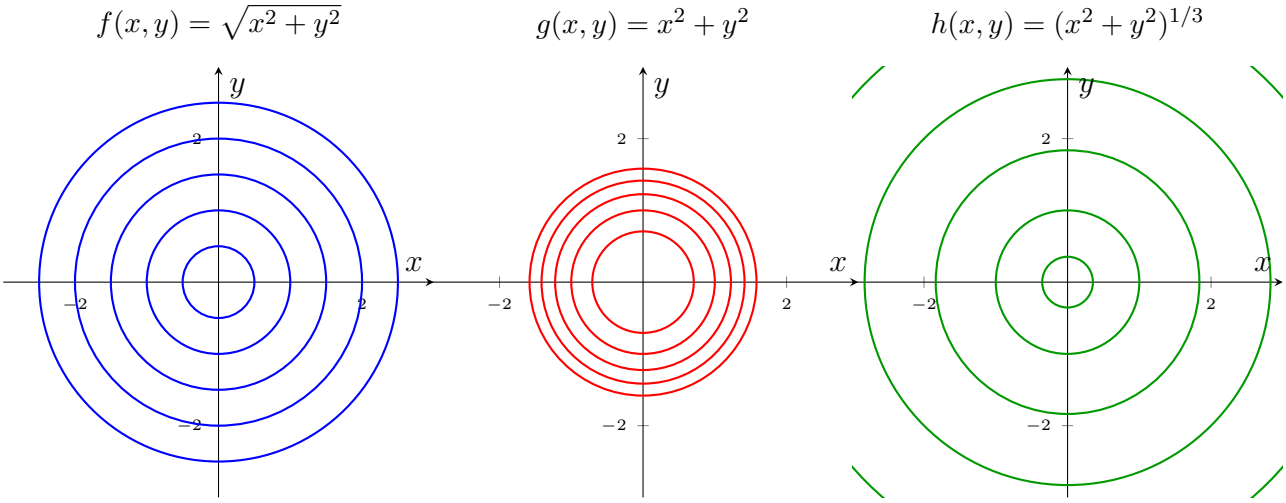
$$\{(x, y) \in \mathbb{R}^2 : f(x, y) = c\}.$$

A collection of such level curves forms a **contour map**.

**Example 8.2** (Comparing Level Curves)**.**

Consider three functions and their level sets $f(x, y) = c$:

1. $f(x, y) = \sqrt{x^2 + y^2}$: Level sets are equally spaced concentric circles $x^2 + y^2 = c^2$ (a cone).
2. $g(x, y) = x^2 + y^2$: Level sets are circles $x^2 + y^2 = c$ spaced closer together as $c$ increases (a paraboloid).
3. $h(x, y) = (x^2 + y^2)^{1/3}$: Level sets are circles $x^2 + y^2 = c^3$ spaced further apart as $c$ increases.




## 8.3 Visualizing Higher Dimensions via Slices

- A sphere in $\mathbb{R}^3$ defined by $x^2 + y^2 + z^2 = \rho^2$ can be sliced along $z = c$ to produce 2D circular cross-sections in $\mathbb{R}^2$.

- A 3-sphere in $\mathbb{R}^4$ defined by $x^2 + y^2 + z^2 + w^2 = R^2$ can be visualized through its 3D cross-sections ($w = c$), which appear as concentric 2-spheres in $\mathbb{R}^3$ whose radii expand from 0 to $R$ and then contract back to 0.

3D Sphere Slices ($z = c$) 4D Hypersphere Slices ($w = c$ in $\mathbb{R}^3$)




## 8.4 Limits of Functions of Several Variables

For single-variable functions, we say that the limit of a function $f(x)$ exists such that

$$\lim_{x \to a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon$$

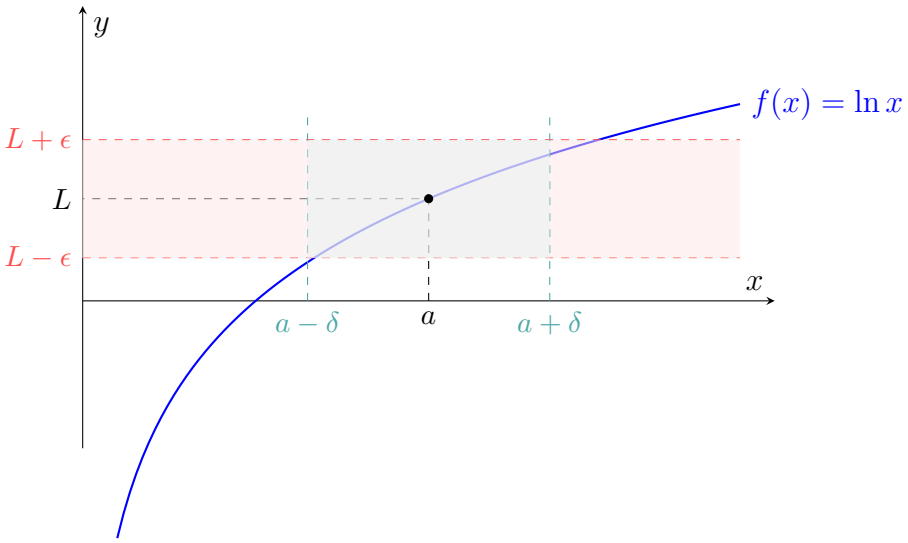
As shown below, we can visualize the $\epsilon$-$\delta$ definition of a limit for a single-variable function $f(x) = \ln x$ at $x = a$.

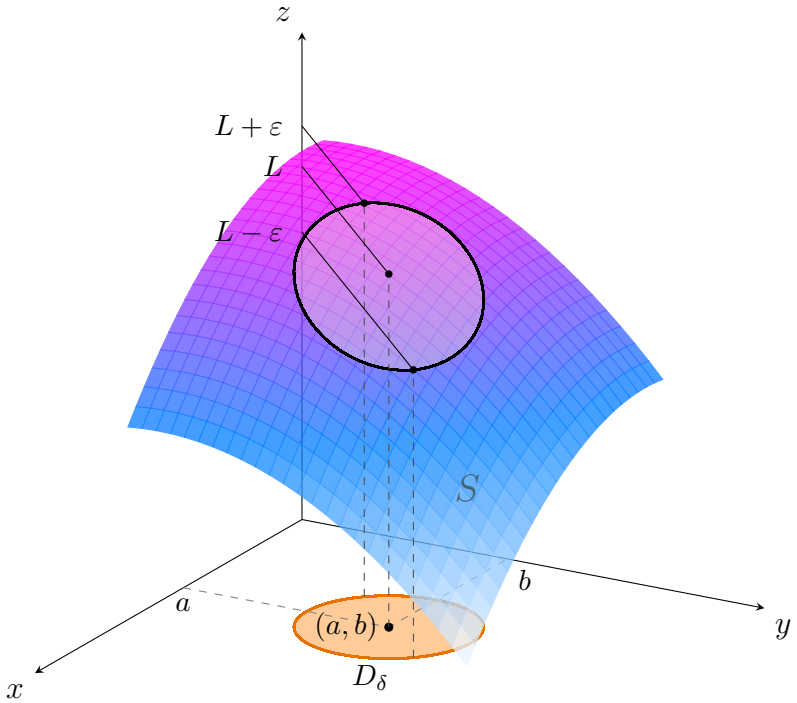
Now consider the case of functions of two variables. We say that the limit of a function $f(x, y)$ exists such that

$$\lim_{(x,y) \to (a,b)} f(x, y) = L \iff \forall \epsilon > 0, \exists \delta > 0 \text{ such that } 0 < \sqrt{(x - a)^2 + (y - b)^2} < \delta$$
$$\implies |f(x, y) - L| < \epsilon$$

Similarly, we can visualize the $\epsilon$-$\delta$ definition of a limit for a two-variable function $f(x, y)$.

$\epsilon$-$\delta$ Definition of Limit for $f(x) = \ln x$ at $x = a$.



## Problem 8.1: Limit of a Function of Two Variables

Find, if it exists, $\lim_{(x,y)\to(0,0)} \dfrac{xy}{x^2+y^2}$.

## Solution 8.1

First, we consider approaching $(0,0)$ along the $x$-axis, i.e., $y=0$:

$$\lim_{x\to 0} \frac{x(0)}{x^2+0^2} = 0$$

Next, we consider approaching $(0,0)$ along the $y$-axis, i.e., $x=0$:

$$\lim_{y\to 0} \frac{0(y)}{0^2+y^2} = 0$$

Next, we consider approaching $(0,0)$ along the line $y=x$:

$$\lim_{x\to 0} \frac{x(x)}{x^2+x^2} = \lim_{x\to 0} \frac{x^2}{2x^2} = \lim_{x\to 0} \frac{1}{2} = \frac{1}{2}$$

Since approaching along different paths yields different limiting values, the limit does not exist.

**Systematic Path Approaches** Checking paths individually can be cumbersome. Instead, we can systematically analyze general path families:

1. Linear Paths ($y=mx$):

$$\lim_{x\to 0} f(x,mx) = \lim_{x\to 0} \frac{x(mx)}{x^2+(mx)^2} = \lim_{x\to 0} \frac{mx^2}{x^2(1+m^2)} = \frac{m}{1+m^2}$$

The limit depends explicitly on the slope $m$. Hence, the limit does not exist.

2. Polar Coordinates ($x=r\cos\theta, y=r\sin\theta$):

$$\lim_{r\to 0} f(r\cos\theta, r\sin\theta) = \lim_{r\to 0} \frac{r^2\cos\theta\sin\theta}{r^2} = \cos\theta\sin\theta$$

The limit depends on the angle $\theta$, confirming the limit does not exist.

**Definition 8.3: Squeeze Theorem for Two Variables**

If $g(x,y) \leq f(x,y) \leq h(x,y)$ for all $(x,y)$ in a disk centered at $(a,b)$ (except possibly at $(a,b)$ itself), and

$$\lim_{(x,y)\to(a,b)} g(x,y) = \lim_{(x,y)\to(a,b)} h(x,y) = L$$

then $\lim_{(x,y)\to(a,b)} f(x,y) = L$.

**Definition 8.4: Continuity of Functions of Several Variables**

A function $f(x,y)$ is *continuous* at $(a,b)$ if:

$$\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$$

Furthermore, $f(x,y)$ is continuous on a domain $D$ if it is continuous at every point $(a,b) \in D$.

**Example 8.3** (Continuity of a Composite Function)**.**
Consider $f(x,y) = \arctan\left(\frac{y}{x}\right)$.

The inner function $g(x,y) = \frac{y}{x}$ is continuous everywhere except on the line $x = 0$. Since $h(t) = \arctan(t)$ is continuous on $\mathbb{R}$, the composite function $f(x,y) = h(g(x,y))$ is continuous at all points $(x,y) \in \mathbb{R}^2$ where $x \neq 0$.

# 9 Partial Derivatives

## 9.1 Definition and Properties of Partial Derivatives

**Definition 9.1: Partial Derivative**

In this course, we will focus on functions of two variables, i.e., $f(x, y)$. For a binary function $f(x, y)$, the partial derivative with respect to $x$ is given by:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

and the partial derivative with respect to $y$ is given by:

$$\frac{\partial f}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

provided these limits exist.

In general, let $f(x_1, x_2, \ldots, x_n)$ be a function of $n$ variables. The **partial derivative** of $f$ with respect to the variable $x_i$ is defined as:

$$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_i + h, \ldots, x_n) - f(x_1, x_2, \ldots, x_i, \ldots, x_n)}{h}$$

provided this limit exists.

Some key points to note about partial derivatives:

- $f_x(a, b)$ is the instantaneous rate of change of $f$ at the point $(a, b)$ in the $x$-direction.
- $f_x(a, b)$ is the slope of the tangent line to $f(a, b)$ at the point $x = a$.

Computing partial derivatives is similar to computing derivatives in single-variable calculus. The only difference is that we treat all other variables as constants when differentiating with respect to one variable.

**Example 9.1** (Partial Derivatives of a Binary Function)**.**

Let $f(x, y) = e^{2x} \cos(y)$.

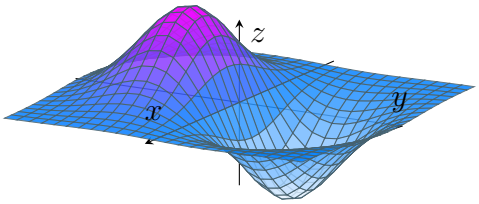
We can now compute the partial derivatives of $f$ with respect to $x$ and $y$:

$$f_x(x,y) = \frac{\partial}{\partial x}\left(e^{2x}\cos(y)\right) = 2e^{2x}\cos(y)$$
$$f_y(x,y) = \frac{\partial}{\partial y}\left(e^{2x}\cos(y)\right) = -e^{2x}\sin(y)$$

For better comprehension, we can visualize the partial derivatives of a function of two variables as the slopes of tangent lines to the surface defined by the function in the $x$ and $y$ directions.

**Example 9.2** (Visualizing Partial Derivatives)**.**

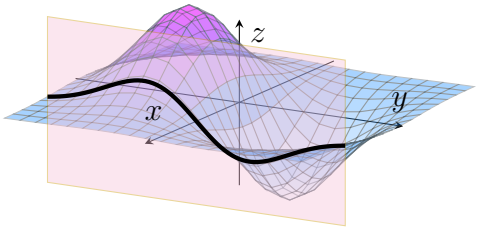
Consider the surface defined by $f(x,y) = -\frac{10y}{e^{x^2+y^2}}$ and its slices along fixed coordinate planes.

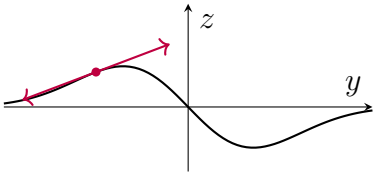



*(Original surface $z = f(x,y)$)*

$x = 1$ **Trace:**

$$z = f(1,y) = -\frac{10y}{e^{1+y^2}}$$

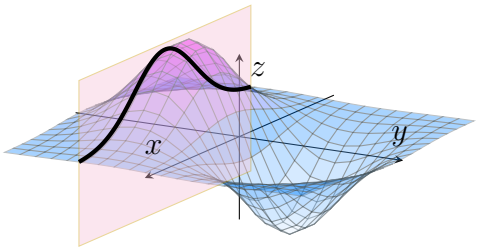


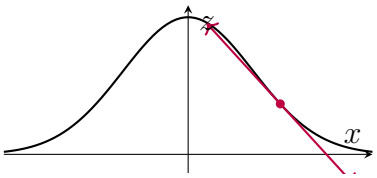

At $(1,-1)$,

$$f_y(1,-1) = \frac{10}{e^2} > 0.$$

$y = -1$ **Trace:**




$$z = f(x, -1) = \frac{10}{e^{x^2+1}}$$

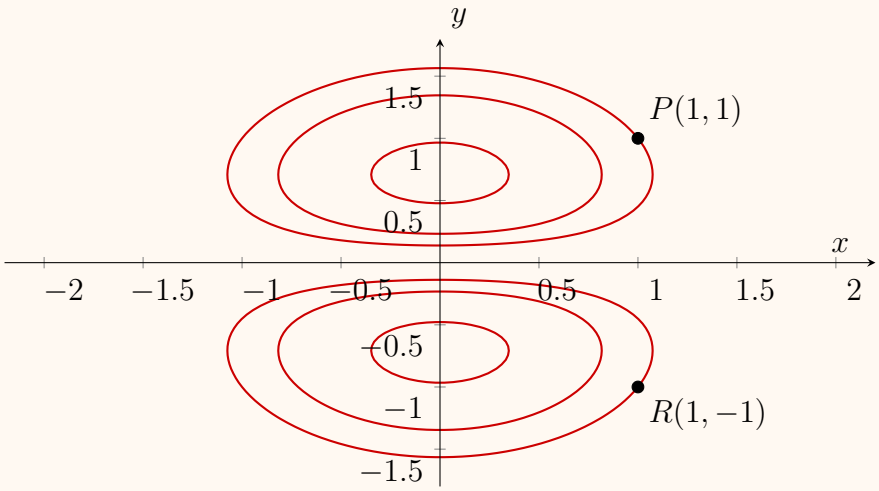



At $(1, -1)$,

$$f_x(1, -1) = -\frac{20}{e^2} < 0.$$

**Problem 9.1: Visualizing Partial Derivatives on a Contour Map**

Give some reasonable labelings to the contour map based on the image of the graph above.




Determine whether the following values are **positive**, **negative**, or **0**:

1. $f_x(O)$ and $f_y(O)$ at the origin $O(0, 0)$.
2. $f_x(P)$ and $f_y(P)$ at $P(1, 1)$.
3. Compare $|f_x(R)|$ and $|f_y(R)|$ at $R(1, -1)$. Which is greater?

**Solution 9.1**

1. At $O(0,0)$:
$$f_x(0,0) = 0, f_y(0,0) = -10 < 0$$
2. At $P(1,1)$: As $x$ increases, $f(x,1)$ increases towards 0 from negative values, so $f_x(P) > 0$. At $y = 1$, $f(1,y)$ reaches a local extremum, so $f_y(P) = 0$.
3. At $R(1,-1)$: The contour lines are significantly denser in the $y$-direction than in the $x$-direction. Thus:
$$|f_x(R)| > |f_y(R)|$$

## 9.2 Linear Approximation

**Definition 9.2: Linear Function**

A **linear function** of two variables is a function of the form

$$f(x,y) = m_x x + m_y y + c$$

The idea of linear approximation is to approximate a function $f(x,y)$ by a linear function $L(x,y)$ that is tangent to the surface defined by $f$ at a given point $(a,b)$.

Given a function $L(x,y) = m_x(x-a) + m_y(y-b) + C$.

Firstly, we want $L(a,b) = f(a,b)$. We then write the linear function as

$$L(x,y) = m_x(x-a) + m_y(y-b) + f(a,b)$$

Secondly, we want the traces of $L$ to be tangent to the traces of $f$ at $(a,b)$. Hence, we have

$$m_x = f_x(a,b), \quad m_y = f_y(a,b)$$

Therefore,

$$L(x,y) = f_x(a,b) \cdot (x-a) + f_y(a,b) \cdot (y-b) + f(a,b)$$

Thus, we now have the following definition of linear approximation:

**Definition 9.3: Linear Approximation**

The **linear approximation** of a function $f(x, y)$ at the point $(a, b)$ is given by

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b),$$

which is the tangent plane.

**Problem 9.2: Finding a Linear Approximation**

Find the linear approximation of $f(x, y) = \frac{x^2}{y^2+1}$ at $(4, 1)$. Use your answer to approximate $f(4.01, 0.98)$.

**Solution 9.2**

First, we find the partial derivatives of $f(x, y) = \frac{x^2}{y^2+1}$:

$$f_x(x, y) = \frac{2x}{y^2 + 1}, \quad f_y(x, y) = -\frac{2x^2y}{(y^2 + 1)^2}$$

At the point $(4, 1)$:

$$f(4, 1) = \frac{16}{2} = 8, \quad f_x(4, 1) = \frac{8}{2} = 4, \quad f_y(4, 1) = -\frac{32}{4} = -8$$

The linear approximation is:

$$z = 8 + 4(x - 4) + (-8)(y - 1)$$

Simplifying:

$$z = 8 + 4x - 16 - 8y + 8 = 4x - 8y$$

To approximate $f(4.01, 0.98)$:

$$f(4.01, 0.98) \approx 4(4.01) - 8(0.98) = 16.04 - 7.84 = 8.2$$

## 9.3 Differentiability

**Definition 9.4: Differentiability**

A function $f(x, y)$ is said to be **differentiable** at a point $(a, b)$ if the linear approximation of $f$ at $(a, b)$ is a good approximation of $f$ near $(a, b)$.

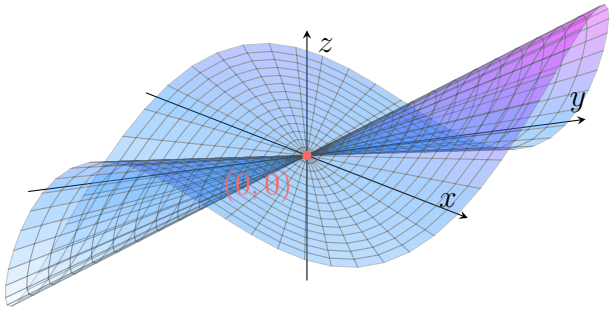
This is equivalent to saying that $f(x, y)$ has a tangent plane at $(a, b)$.

A formal definition of differentiability is as follows:

$$\lim_{(x,y)\to(a,b)} \frac{f(x, y) - L(x, y)}{\sqrt{(x - a)^2 + (y - b)^2}} = 0$$

**Example 9.3** (**Non-example:**).

$$g(x, y) = \begin{cases} \dfrac{2xy(x + y)}{x^2 + y^2} & (x, y) \neq (0, 0) \\ \\ 0 & (x, y) = (0, 0) \end{cases}.$$




Note that in single variable calculus, if a function has a derivative at a point, it is also differentiable at that point. However, in multivariable calculus, this is not necessarily the case.

This function $g(x, y)$ is a tricky function. From calculation, we know its partial derivatives,

$\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$, exist at $(0,0)$:

$$\begin{aligned} g_x(0,0) &= \lim_{h\to 0} \frac{g(h,0)-g(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \\ g_y(0,0) &= \lim_{h\to 0} \frac{g(0,h)-g(0,0)}{h} \\ &= \lim_{h\to 0} \frac{0-0}{h} = 0 \end{aligned}$$

In fact, from all directions, the derivatives of $g$ exists.

However, $g$ is not **differentiable** at $(0,0)$.

To show this, we can use the definition of differentiability.

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{g(x,y)-L(x,y)}{\sqrt{x^2+y^2}} &= \lim_{(x,y)\to(0,0)} \frac{g(x,y)-g(0,0)+g_x(0,0)(x-0)+g_y(0,0)(y-0)}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{g(x,y)}{\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)\sqrt{x^2+y^2}} \\ &= \lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)^{3/2}} \end{aligned}$$

We approach $(0,0)$ along the line $y = x$:

$$\begin{aligned}
\lim_{(x,y)\to(0,0)} \frac{2xy(x+y)}{(x^2+y^2)^{3/2}} &= \lim_{x\to 0} \frac{2x \cdot x(x+x)}{(x^2+x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{2x^2(2x)}{(2x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{4x^3}{(2x^2)^{3/2}} \\
&= \lim_{x\to 0} \frac{4x^3}{2^{3/2}x^3} \\
&= \lim_{x\to 0} \frac{4}{2^{3/2}} \\
&= \frac{4}{2\sqrt{2}} = \sqrt{2} \neq 0
\end{aligned}$$

This shows that linear approximation is not a good approximation of $g$ near $(0,0)$, and hence $g$ is not differentiable at $(0,0)$.

We now introduce a theorem that provides a sufficient condition for differentiability.

**Theorem 9.1: Sufficient Condition for Differentiability**

If $f_x$ and $f_y$ are continuous on an open disk $D$, then $f(x,y)$ is differentiable on $D$.

## 9.4 Higher-Order Partial Derivatives

**Definition 9.5: Higher-Order Partial Derivatives**

We denote by $f_{xx}$ the function

$$f_{xx}(x,y) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial f}{\partial x}$$

and similarly for $f_{yy}$.

We denote by $f_{xy}$ the function

$$f_{xy}(x,y) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x \partial y}$$

and similarly for $f_{yx}$.

**Note that the order of differentiation matters for mixed partial derivatives.**

The interpretation of higher-order partial derivatives is similar to that of first-order partial derivatives.

- $f_{xx}(a,b)$ is the instantaneous rate of change of $f_x$ at the point $(a,b)$ in the $x$-direction.
- $f_{xy}(a,b)$ is the instantaneous rate of change of $f_x$ at the point $(a,b)$ in the $y$-direction.
- $f_{yx}(a,b)$ is the instantaneous rate of change of $f_y$ at the point $(a,b)$ in the $x$-direction.

**Theorem 9.2: Clairaut's Theorem**

Suppose that $f_{xy}$ and $f_{yx}$ are continuous on some open disk $D$. Then

$$f_{xy}(a,b) = f_{yx}(a,b) \quad \text{for all } (a,b) \in D$$

It is similar for functions of more than two variables.

**Example 9.4** (Application of Clairaut's Theorem)**.**
Let $g(x,y,z,w) = x^3w^2z^2 + \sin\left(\frac{xy}{z^2}\right)$. Now we want to find $g_{zzwx}$.

First, we examine the function $g$. Notice that the first term, $x^3w^2z^2$, is a polynomial function, which is infinitely differentiable over $\mathbb{R}^n$.

The second term, $\sin\left(\frac{xy}{z^2}\right)$, is a composite function. However, sin is infinitely differentiable over $\mathbb{R}$, and $\frac{xy}{z^2}$ also infinitely differentiable over $\mathbb{R}^n, z \neq 0$.

Hence, $g$ is infinitely differentiable over $\mathbb{R}^n, z \neq 0$.

This gives that

$$g_{zzwx} = \frac{\partial g}{\partial z, z, w, x} = g_{wzzx} = g_{xwzz} = \cdots$$

If we calculate this in the original order, we have

$$\begin{aligned} g_{zzwx} &= (((g_z)_z)_w)_x \\ &= (((6x^3w^2z - \frac{2y}{z^3}\cos\left(\frac{xy}{z^2}\right) - \frac{2xy^2}{z^5}\sin\left(\frac{xy}{z^2}\right))_z)_w)_x \\ &= (((6x^3w^2 + \frac{6y}{z^4}\cos\left(\frac{xy}{z^2}\right) + \frac{10xy^2}{z^6}\sin\left(\frac{xy}{z^2}\right) - \frac{4x^2y^3}{z^8}\cos\left(\frac{xy}{z^2}\right))_w)_x \\ &= \cdots \end{aligned}$$

We want to try to avoid this tedious calculation by using Clairaut's Theorem. Rearranging the order of differentiation to $g_{xwzz}$, we have

$$\begin{aligned} g_{xwzz} &= (((g_x)_w)_z)_z \\ &= (((3x^2w^2z^2 + \frac{y}{z^2}\cos\left(\frac{xy}{z^2}\right))_w)_z)_z \\ &= (((6x^2wz^2 + 0)_z)_z) \\ &= ((12x^2wz + 0)_z) \\ &= 12x^2w \end{aligned}$$

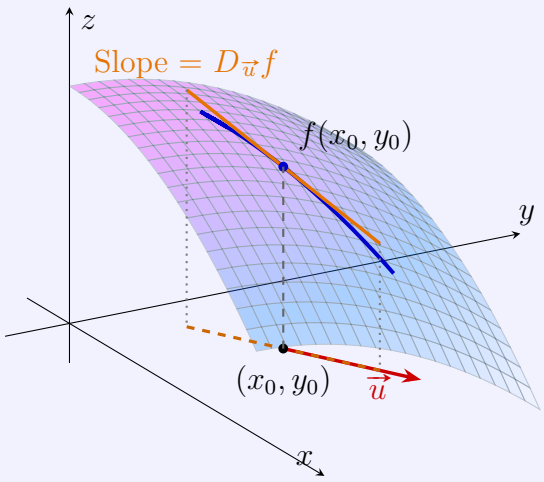
## 9.5 Directional Derivatives

**Definition 9.6: Directional Derivative**

Let $\vec{u}$ be a **unit vector**, and let $f(x, y)$ be a function of two variables. The **directional derivative** of $f$ at the point $(a, b)$ in the direction of $\vec{u}$ is defined as

$$D_{\vec{u}}f(\vec{r_0}) = \lim_{t\to 0}\frac{f(\vec{r_0} + t\vec{u}) - f(\vec{r_0})}{t}$$

Here, we visualize the directional derivative as the slope of the tangent line to the curve on the surface defined by $f$ in the direction of $\vec{u}$ at the point $(a, b)$.




**Example 9.5** (Directional Derivative of a Function of Two Variables)**.**

Let $f(x, y) = ax + by$, $\vec{u} = \langle \cos\theta, \sin\theta \rangle$. We now want to find $D_{\vec{u}} f$.

$$
\begin{aligned}
D_{\vec{u}} f &= \lim_{t \to 0} \frac{f(\vec{r_0} + t\vec{u}) - f(\vec{r_0})}{t} \\
&= \lim_{t \to 0} \frac{f(x_0 + t\cos\theta, y_0 + t\sin\theta) - f(x_0, y_0)}{t} \\
&= \lim_{t \to 0} \frac{a(x_0 + t\cos\theta) + b(y_0 + t\sin\theta) - (ax_0 + by_0)}{t} \\
&= \lim_{t \to 0} \frac{at\cos\theta + bt\sin\theta}{t} \\
&= a\cos\theta + b\sin\theta \\
&= \vec{u} \cdot \langle a, b \rangle
\end{aligned}
$$

## 9.6 Gradient

**Definition 9.7: Gradient**

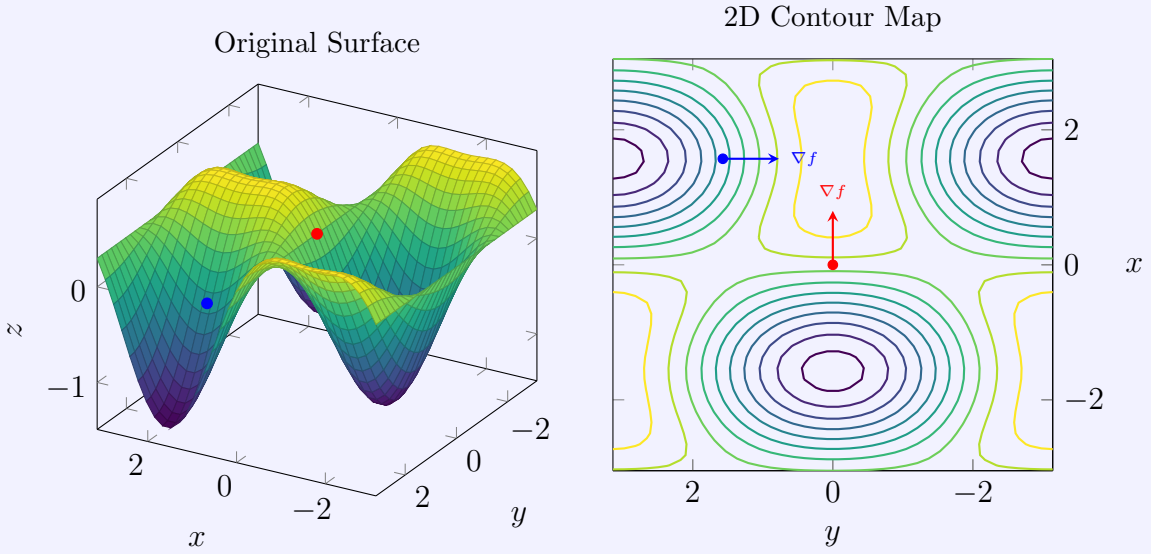
The **gradient** of a function $f(x, y)$ is the vector $\langle a, b \rangle$ such that

$$\langle a, b \rangle = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

We call this vector the **gradient vector** of $f$ and denote it by $\nabla f$.

Here, we visualize the gradient vector as a vector that points in the direc-

tion of the steepest ascent of the surface, as shown below in the graphs.




**Theorem 9.3: Gradient Theorem**

If $f$ is differentiable at $(a, b)$, then

$$\begin{aligned} D_{\vec{u}} f &= \nabla f \cdot \vec{u} \\ &= \|\nabla f\| \cos\theta \end{aligned}$$

Given this, we know that:

- $\nabla f$ points in the direction of the **fastest increase** of $f$.
- $-\nabla f$ points in the direction of the **fastest decrease** of $f$.
- $\nabla f$ is **perpendicular** to the level curves of $f$.

**Example 9.6** (Gradient of a Function of Two Variables)**.**

Let $f(x, y) = \sin(xy^2)$. We have

$$\nabla f = \langle f_x, f_y \rangle = \left\langle y^2 \cos\left(xy^2\right), 2xy \cos\left(xy^2\right) \right\rangle$$

Given this, we can find the gradient of $f$ at $(0, 1)$.

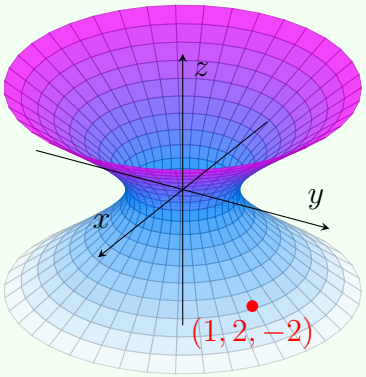
$$\nabla f_{(0,1)} = \langle 1, 0 \rangle$$

## Problem 9.3: Finding the Tangent Plane of a Surface

Let $f(x, y, z) = x^2 + y^2 - z^2$, and the surface be defined by $f(x, y, z) = 1$.

Find the tangent plane of the surface at the point $(1, 2, -2)$.

## Solution 9.3




First, we compute the gradient of $f$:

$$\nabla f = \langle f_x, f_y, f_z \rangle = \langle 2x, 2y, -2z \rangle$$

Next, we evaluate the gradient at the point $(1, 2, -2)$:

$$\nabla f(1, 2, -2) = \langle 2, 4, 4 \rangle$$

The equation of the tangent plane at the point $(x_0, y_0, z_0)$ is given by:

$$\nabla f(x_0, y_0, z_0) \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

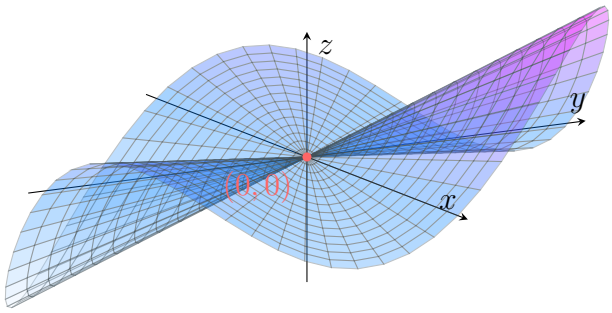
Substituting the values, we have:

$$\begin{aligned}
\langle 2, 4, 4\rangle \cdot \langle x-1, y-2, z+2\rangle &= 0 \\
2(x-1) + 4(y-2) + 4(z+2) &= 0 \\
2x - 2 + 4y - 8 + 4z + 8 &= 0 \\
2x + 4y + 4z - 2 &= 0 \\
x + 2y + 2z - 1 &= 0
\end{aligned}$$

## 9.7 Differentiability and Continuity

Note that it is dangerous to assume that the existence of partial derivatives implies differentiability.

Recall the example where we proved that the function $g(x, y) = \dfrac{2xy(x+y)}{x^2+y^2}$ is not differentiable at the origin, even though its partial derivatives exist there.



Again, we can compute the gradient of $g$ at $(0,0)$, with $\vec{u} = \langle \cos\theta, \sin\theta \rangle$.

$$\begin{aligned}
D_{\vec{u}} g &= \lim_{t\to 0} \frac{g(\vec{r_0} + t\vec{u}) - g(\vec{r_0})}{t} \\
&= \lim_{t\to 0} \frac{g(t\cos\theta, t\sin\theta) - g(0,0)}{t} \\
&= \lim_{t\to 0} \frac{\dfrac{2(t\cos\theta)(t\sin\theta)(t\cos\theta + t\sin\theta)}{(t\cos\theta)^2 + (t\sin\theta)^2}}{t} \\
&= \lim_{t\to 0} \frac{\dfrac{2t^3(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta)}{t^2(\cos^2\theta + \sin^2\theta)}}{t} \\
&= \lim_{t\to 0} \frac{2t^3(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta)}{t^3} \\
&= \lim_{t\to 0} 2(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta) \\
&= 2(\cos^2\theta\sin\theta + \cos\theta\sin^2\theta) \\
&= 2\cos\theta\sin\theta(\cos\theta + \sin\theta)
\end{aligned}$$

This result is not a constant, and it depends on the direction $\vec{u}$. It yields 0 when $\theta = k\pi/2, k \in \mathbb{Z}$, and $\pm 1$ when $\theta = \pi/4 + k\pi, k \in \mathbb{Z}$.

Note that when

$$D_{\vec{u}} f(x,y) \neq \nabla f \cdot \vec{u}$$

The function is not 'behaving like a plane' at the point $(x,y)$, and hence it is not differentiable at that point.

**Theorem 9.4: Differentiability and Continuity**

If $f_x$ and $f_y$ are continuous on an open disk $D$, then $f$ is differentiable on $D$.

## 9.8 Chain Rule for Multivariable Functions

**Example 9.7** (A Practice Problem)**.**

Given differentiable functions $\vec{r}(t) = \langle x(t), y(t) \rangle$ and $f(x, y)$.

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t)) = \lim_{h \to 0} \frac{f(\vec{r}(t+h)) - f(\vec{r}(t))}{h}$$

We now compute the linear approximation of $f$ at $\vec{r}(t+h)$:

$$\begin{aligned} f(\vec{r}(t+h)) &\approx f(\vec{r}(t)) + f_x(\vec{r}(t))(x(t+h) - x(t)) + f_y(\vec{r}(t))(y(t+h) - y(t)) \\ &= f(\vec{r}(t)) + f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y, \\ \text{where } \Delta x &= x(t+h) - x(t), \\ \Delta y &= y(t+h) - y(t) \end{aligned}$$

Hence, the original limit becomes

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} f(\vec{r}(t)) &= \lim_{h \to 0} \frac{f(\vec{r}(t)) + f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y - f(\vec{r}(t))}{h} \\ &= \lim_{h \to 0} \frac{f_x(\vec{r}(t))\Delta x + f_y(\vec{r}(t))\Delta y}{h} \\ &= \lim_{h \to 0} \left( f_x(\vec{r}(t)) \frac{\Delta x}{h} + f_y(\vec{r}(t)) \frac{\Delta y}{h} \right) \\ &= f_x(\vec{r}(t)) \lim_{h \to 0} \frac{\Delta x}{h} + f_y(\vec{r}(t)) \lim_{h \to 0} \frac{\Delta y}{h} \\ &= f_x(\vec{r}(t)) \frac{\mathrm{d}}{\mathrm{d}t} x(t) + f_y(\vec{r}(t)) \frac{\mathrm{d}}{\mathrm{d}t} y(t) \end{aligned}$$

From this, we can derive the Chain Rule for multivariable functions.

**Theorem 9.5: Chain Rule for Two-Variable Functions**

Let $f(x, y)$ be a differentiable function of two variables, and let $\vec{r}(t) = \langle x(t), y(t) \rangle$ be

a differentiable vector-valued function of one variable. Then

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) &= f_x(\vec{r}(t))x'(t) + f_y(\vec{r}(t))y'(t)\\ &= \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)\\ &= \frac{\partial f}{\partial x}\frac{\mathrm{d}}{\mathrm{d}t}x + \frac{\partial f}{\partial y}\frac{\mathrm{d}}{\mathrm{d}t}y\end{aligned}$$

If we examine it from basic concepts, we can see how the Chain Rule naturally holds.

$$\begin{aligned}\frac{\Delta f}{\Delta t} &= \frac{\frac{\Delta f}{\Delta x}\Delta x + \frac{\Delta f}{\Delta y}\Delta y}{\Delta t}\\ &= \frac{\Delta f}{\Delta x}\frac{\Delta x}{\Delta t} + \frac{\Delta f}{\Delta y}\frac{\Delta y}{\Delta t}\end{aligned}$$

From the Chain Rule and the definition of directional derivative, we can derive the following theorem.

**Theorem 9.6: Directional Derivative and Gradient**

Let $f(x, y)$ be a differentiable function of two variables, and let $\vec{u}$ be a unit vector. Then, $\langle u_1, u_2, D_{\vec{u}}f\rangle$ is tangent to $f$.

Recall that $D_{\vec{u}}f \perp$ the level curves of $f$. But why is that? Let's explore this concept further.

If $\vec{r}$ follows a level curve of $f$, then $f(\vec{r}(t))$ is constant. Hence, $\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = 0$. Meanwhile, $\frac{\mathrm{d}}{\mathrm{d}t}f(\vec{r}(t)) = \nabla f(\vec{r}(t)) \cdot \vec{r}'(t)$.

Therefore, we have

$$\begin{aligned}&\nabla f(\vec{r}(t)) \cdot \vec{r}'(t) = 0\\ \Longrightarrow\ &\nabla f(\vec{r}(t)) \perp \vec{r}'(t)\end{aligned}$$

**Problem 9.4: Partial Derivatives of Composite Functions**

Given $x = r\cos\theta, y = r\sin\theta, f(x,y) = x^2 + y^2$, find $\dfrac{\partial f}{\partial r}$ and $\dfrac{\partial f}{\partial \theta}$.

**Solution 9.4**

We can use the Chain Rule to find the partial derivatives of $f$ with respect to $r$ and $\theta$.

$$\begin{aligned}
\frac{\partial f}{\partial r} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} \\
&= 2x\cos\theta + 2y\sin\theta \\
&= 2(r\cos\theta)\cos\theta + 2(r\sin\theta)\sin\theta \\
&= 2r(\cos^2\theta + \sin^2\theta) \\
&= 2r
\end{aligned}$$

Similarly, we can find $\frac{\partial f}{\partial \theta}$:

$$\begin{aligned}
\frac{\partial f}{\partial \theta} &= \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} \\
&= 2x(-r\sin\theta) - 2y(r\cos\theta) \\
&= -2rx\sin\theta - 2ry\cos\theta \\
&= -2r(r\cos\theta)\sin\theta - 2r(r\sin\theta)\cos\theta \\
&= -2r^2\cos\theta\sin\theta - 2r^2\sin\theta\cos\theta \\
&= -4r^2\cos\theta\sin\theta
\end{aligned}$$

Hereby, we have found the partial derivative for $f$ without directly substituting $x$ and $y$ in terms of $r$ and $\theta$.

**Theorem 9.7: Chain Rule for Multivariable Functions**

Let $f(x_1, x_2, \ldots, x_n)$ be a differentiable function of $n$ variables, and let $x_i =$

$x_i(t_1, t_2, \ldots, t_m)$ for $i = 1, 2, \ldots, n$. Then,

$$\begin{aligned}\frac{\partial f}{\partial t_j} &= \frac{\partial f}{\partial x_1}\frac{\partial x_1}{\partial t_j} + \frac{\partial f}{\partial x_2}\frac{\partial x_2}{\partial t_j} + \cdots + \frac{\partial f}{\partial x_n}\frac{\partial x_n}{\partial t_j} \\ &= \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\frac{\partial x_i}{\partial t_j}, \quad j = 1, 2, \ldots, m.\end{aligned}$$

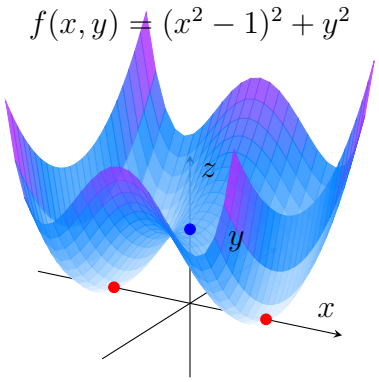
# 10 Optimization and Lagrange Multipliers

## 10.1 Extrema of Multivariable Functions

**Definition 10.1: Local Maximum and Minimum**

For a function $f(x, y)$ and a point $(a, b)$ in its domain, we say that $(a, b)$ is a

- **local maximum** if $f(a, b) \geq f(x, y),\ \forall (x, y)$ in a neighborhood of $(a, b)$.
- **local minimum** if $f(a, b) \leq f(x, y),\ \forall (x, y)$ in a neighborhood of $(a, b)$.




**Definition 10.2: Critical Point**

We call a point $P(a, b)$ a **critical point** of $f$ if

$$\nabla f(P) = \vec{0}$$

**Theorem 10.1: Critical Points and Local Extrema**

If $P(a, b)$ is a local extremum of $f$, then $P$ is a critical point of $f$.

For single variable functions, we can use the second derivative test to determine the nature of a point on the function. For $f(x)$, if $f'(a) = 0$, then:

- If $f''(a) > 0$, then $f$ has a local minimum at $x = a$.
- If $f''(a) < 0$, then $f$ has a local maximum at $x = a$.
- If $f''(a) = 0$, the test is inconclusive.

For multivariable functions, we can use its quadratic approximation to determine the nature

of a critical point.

**Problem 10.1: Determining the Critical Points of a Function**

Find critical points of the function $f(x, y) = x^4 + y^4 + 4xy$. Which are local maxima, local minima, and saddle points?

**Solution 10.1**

First, we find the critical points by setting the gradient of $f$ to zero. The gradient is given by:

$$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle = (4x^3 + 4y, 4y^3 + 4x)$$

Setting each component to zero gives us the system of equations:

$$4x^3 + 4y = 0$$
$$4y^3 + 4x = 0$$

This implies $y = -x^3$. Substituting $y = -x^3$ into the second equation gives:

$$4(-x^3)^3 + 4x = 0 \implies -4x^9 + 4x = 0 \implies 4x(-x^8 + 1) = 0$$

This gives us $x = 0$ or $x^8 = 1$. The equation $x^8 = 1$ has solutions $x = \pm 1$. So the critical points are $(0, 0)$, $(1, -1)$, and $(-1, 1)$.

To determine the nature of these critical points, we write $f(x, y)$ in the quadratic approximation form around each critical point.

$$f(x, y) \approx a + bx + cy + Ax^2 + Bxy + Cy^2$$
$$\text{If } \nabla f(0, 0) = 0, \text{ then } b = c = 0, a = f(0, 0)$$
$$\implies f(x, y) \approx f(0, 0) + Ax^2 + Bxy + Cy^2$$

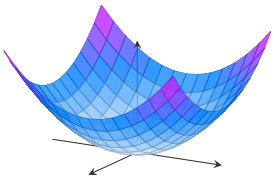
The second derivative is then given by

$$\frac{\partial^2 f}{\partial x \partial x} = 2A$$
$$\frac{\partial^2 f}{\partial y \partial y} = 2C$$
$$\frac{\partial^2 f}{\partial x \partial y} = B$$

Fact: If $(a, b)$ is a critical point of $f$, then the quadratic approximation is given by

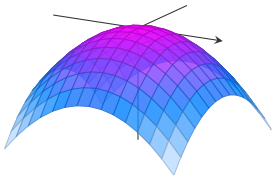
$$Q(x, y) = \frac{1}{2} f_{xx}(a, b)(x - a)^2 + f_{xy}(a, b)(x - a)(y - b) + \frac{1}{2} f_{yy}(a, b)(y - b)^2$$

There are three possible nondegenerate behaviors for a quadratic function near a critical point:



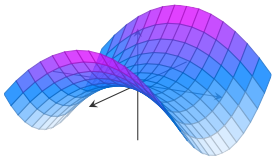
$Q(x, y) = x^2 + y^2$

**Minimum**



$Q(x, y) = -x^2 - y^2$

**Maximum**



$Q(x, y) = x^2 - y^2$

**Saddle**

For a general quadratic function

$$Q(x, y) = Ax^2 + Bxy + Cy^2,$$

its behavior is determined by $4AC - B^2$. For the quadratic approximation of $f$ at a critical point $(a, b)$,

$$A = \frac{1}{2} f_{xx}(a, b), \qquad B = f_{xy}(a, b), \qquad C = \frac{1}{2} f_{yy}(a, b).$$

Hence,

$$4AC - B^2 = f_{xx}(a, b) f_{yy}(a, b) - f_{xy}(a, b)^2.$$

**Definition 10.3: Second Derivative Test**

Let $(a, b)$ be a critical point of $f$, and define

$$D = f_{xx}(a, b)f_{yy}(a, b) - f_{xy}(a, b)^2.$$

Then:

- If $D > 0$ and $f_{xx}(a, b) > 0$, then $(a, b)$ is a **local minimum**.
- If $D > 0$ and $f_{xx}(a, b) < 0$, then $(a, b)$ is a **local maximum**.
- If $D < 0$, then $(a, b)$ is a **saddle point**.
- If $D = 0$, the test is **inconclusive**.

**Example 10.1** (Usage of the Discriminant)**.**

We have $f(x, y) = x^2 + y^2 - 3xy$.

From calculation, we know that $\nabla f = \langle 2x - 3y, 2y - 3x\rangle$. Setting this equal to zero gives us the system of equations:

$$2x - 3y = 0 \quad \text{and} \quad 2y - 3x = 0.$$

Solving this system, we find that the only critical point is $(0, 0)$.

The discriminant is given by $D = (2)(2) - (-3)^2 = 4 - 9 = -5 < 0$.

We now know that $(0, 0)$ is a saddle.

**Definition 10.4: Global Extrema**

Given a function $f(x, y)$ defined on some domain $\mathcal{D}$, we say that $(a, b) \in \mathcal{D}$ is a

- **global maximum** if $f(a, b) \geq f(x, y),\ \forall (x, y) \in \mathcal{D}$.
- **global minimum** if $f(a, b) \leq f(x, y),\ \forall (x, y) \in \mathcal{D}$.

## 10.2 Domains and Boundaries

**Definition 10.5: Interior and Boundary Points**

Let $\mathcal{D} \subseteq \mathbb{R}^2$ be a domain.

- A point $(a, b) \in \mathcal{D}$ is an **interior point** if is has some disk around it in $\mathcal{D}$.
- A point $(a, b) \in \mathcal{D}$ is a **boundary point** such that every disk around it intersects $\mathcal{D}$ and $\mathbb{R}^2 \setminus \mathcal{D}$.

* $\mathbb{R}^2 \setminus D$ means the set of points in $\mathbb{R}^2$ that are not in $D$.

**Definition 10.6: Closed and Bounded Sets**

A set $\mathcal{D} \subseteq \mathbb{R}^2$ is **closed** if it contains all of its boundary points, and **bounded** if it can be contained in a disk of finite radius.

**Theorem 10.2: Extreme Value Theorem**

Let $f$ be a continuous function on a closed and bounded set $\mathcal{D} \subseteq \mathbb{R}^2$.

Then $f$ attains both a global maximum and a global minimum on $\mathcal{D}$.

**Example 10.2** (Finding Global Extrema)**.**

Given the function $f(x, y) = x^2 + y^2 + xy$, and the domain $D = \{(x, y) : x^2 + y^2 \leq 4\}$.

We set the gradient of $f$ to zero to find critical points:

$$\nabla f = \langle 2x + y, 2y + x \rangle = \langle 0, 0 \rangle .$$

This gives us the system of equations:

$$2x + y = 0 \quad \text{and} \quad 2y + x = 0.$$

Solving this system, we find that the only critical point is $(0, 0)$.

Since $\mathcal{D}$ is closed and bounded, and $f$ is continuous on $\mathcal{D}$, the Extreme Value Theorem guarantees that $f$ attains both a global maximum and a global minimum on $\mathcal{D}$. Therefore, we need to check the boundary of $\mathcal{D}$ to find the global extrema.

The boundary of $\mathcal{D}$ is the circle defined by $x^2 + y^2 = 4$. We can parameterize this boundary

using polar coordinates:

$$x = 2\cos(t), \quad y = 2\sin(t), \quad 0 \leq t < 2\pi.$$

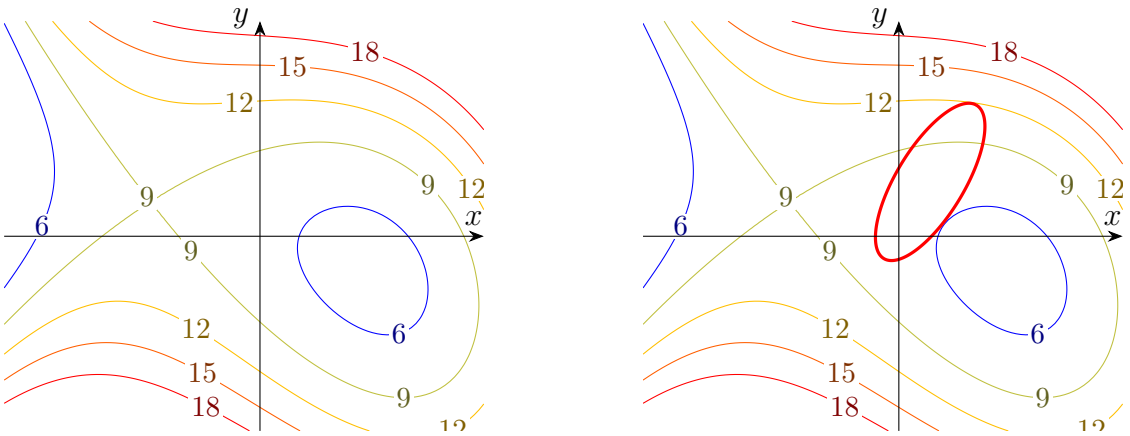
Hence,

$$\begin{aligned} f(2\cos(t), 2\sin(t)) &= (2\cos(t))^2 + (2\sin(t))^2 + (2\cos(t))(2\sin(t)) \\ &= 4\cos^2(t) + 4\sin^2(t) + 4\cos(t)\sin(t) \\ &= 4 + 4\cos(t)\sin(t) \\ &= 4 + 2\sin(2t). \end{aligned}$$

Therefore, the maximum value of $f$ on the boundary is $4 + 2 = 6$ at $x = 2, y = 0$, and the minimum value is $4 - 2 = 2$ at $x = -2, y = 0$.

## 10.3 Lagrange Multipliers

**Example 10.3** (Extrema of a Function on a Curve)**.**
Below is a function $f$ and a curve $C$ given by $g(x, y) = c$.




In this example, the extrema under constraints occur where the level curves of $f$ are tangent to the constraint curve $C$. This is a particularly lucky situation, as we can visually identify the points of tangency.

**Theorem 10.3: Lagrange Multipliers**

Let $f(x, y)$ be a function and $g(x, y) = c$ be a constraint curve. If $g(x, y) = c$ is tangent to a level curve of $f$, then

$$\nabla f \parallel \nabla g \implies \nabla f = \lambda \nabla g$$

Therefore, to find the extrema of $f$ under the constraint $g(x, y) = c$, we can solve the system of equations:

$$\nabla f = \lambda \nabla g, \quad g(x, y) = c$$

for $x$, $y$, and $\lambda$, assuming that $\nabla g \neq 0$ at the points of interest.

**Example 10.4** (Using Lagrange Multipliers)**.**

We want to find the extrema of $f(x, y) = x + 2y$ subject to the constraint $3x^2 + 2y^2 = 7$.

First, we compute the gradients:

$$\nabla f = \langle 1, 2 \rangle, \quad \nabla g = \langle 6x, 4y \rangle.$$

Setting $\nabla f = \lambda \nabla g$ gives us the system of equations:

$$\begin{cases} 1 = \lambda(6x) \\ 2 = \lambda(4y) \\ 3x^2 + 2y^2 = 7 \end{cases}$$

Solving the first two equations for $\lambda$, we get:

$$\begin{cases} \lambda = \dfrac{1}{6x} \\ \lambda = \dfrac{1}{2y} \end{cases}$$

Setting these equal gives us:

$$\frac{1}{6x} = \frac{1}{2y} \implies 2y = 6x \implies y = 3x.$$

Substituting $y = 3x$ into the constraint equation gives:

$$\begin{aligned} 3x^2 + 2(3x)^2 &= 7 \\ 3x^2 + 18x^2 &= 7 \\ 21x^2 &= 7 \\ x^2 &= \frac{1}{3} \\ x &= \pm\frac{1}{\sqrt{3}}. \end{aligned}$$

Thus, the corresponding $y$ values are:

$$y = 3\left(\pm\frac{1}{\sqrt{3}}\right) = \pm\sqrt{3}.$$

Therefore, the points of interest are:

$$\left(\frac{1}{\sqrt{3}}, \sqrt{3}\right) \text{ and } \left(-\frac{1}{\sqrt{3}}, -\sqrt{3}\right).$$

**Example 10.5** (Another Example of Lagrange Multipliers)**.**
Given a function $Q(K, L) = AK^{\alpha}L^{1-\alpha}$. This describes the output when $K$ units of capital and $L$ units of labor are used, where $A > 0$ is a constant and $0 < \alpha < 1$.

Suppose that $A = 1$ and $\alpha = \frac{3}{5}$. Given a budget of \$12 with the cost of capital being \$3 per unit and the cost of labor being \$4 per unit, we want to maximize $Q(K, L)$ subject to the constraint $2K + 3L = 12$.

The process is similar to the previous example. We first rewrite the constraint as $g(K, L) = 2K + 3L - 12 = 0$. Then, we compute the gradients:

$$\begin{aligned} \nabla Q &= \left\langle \frac{3}{5}K^{-2/5}L^{2/5}, \frac{2}{5}K^{3/5}L^{-3/5} \right\rangle \\ \nabla g &= \langle 2, 3 \rangle. \end{aligned}$$

Setting $\nabla Q = \lambda \nabla g$ gives us the system of equations:

$$\begin{cases} \frac{3}{5}K^{-2/5}L^{2/5} = 2\lambda \\ \frac{2}{5}K^{3/5}L^{-3/5} = 3\lambda \\ 2K + 3L = 12 \end{cases}$$

Solving the first two equations for $\lambda$, we get:

$$\frac{3}{10}K^{-2/5}L^{2/5} = \frac{2}{15}K^{3/5}L^{-3/5} \implies 9L = 4K \implies L = \frac{4}{9}K.$$

Substituting $L = \frac{4}{9}K$ into the constraint equation gives:

$$\begin{aligned} 2K + 3\left(\frac{4}{9}K\right) &= 12 \\ 2K + \frac{4}{3}K &= 12 \\ \frac{10}{3}K &= 12 \\ K &= \frac{36}{10} = 3.6. \end{aligned}$$

Then, substituting $K = 3.6$ into $L = \frac{4}{9}K$ gives:

$$L = \frac{4}{9} \cdot 3.6 = 1.6.$$

Therefore, the optimal allocation of resources is $K = 3.6$ units of capital and $L = 1.6$ units of labor.

# 11 Double Integrals

**Definition 11.1: Imprecise Definition of Double Integral**

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the $xy$-plane. The **double integral** is defined by

$$\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$$

where $\mathrm{d}A$ is the area element in the $xy$-plane.

This integral can be interpreted as the volume under the surface $z = f(x, y)$ and above the region $\mathcal{D}$ in the $xy$-plane.

Below are some properties of double integrals that will be useful in evaluating them.

- $\iint_{\mathcal{D}} f(x, y) + g(x, y)\,\mathrm{d}A = \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A + \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$
- $\iint_{\mathcal{D}} c \cdot f(x, y)\,\mathrm{d}A = c \cdot \iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A$
- If $f(x, y) \leq g(x, y)$ for all $(x, y) \in \mathcal{D}$, then $\iint_{\mathcal{D}} f(x, y)\,\mathrm{d}A \leq \iint_{\mathcal{D}} g(x, y)\,\mathrm{d}A$.

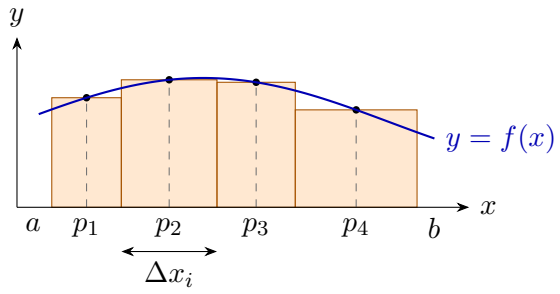
## 11.1 Riemann Sums and the Definition of Double Integrals

Recall the concept of Riemann sums for single-variable functions.

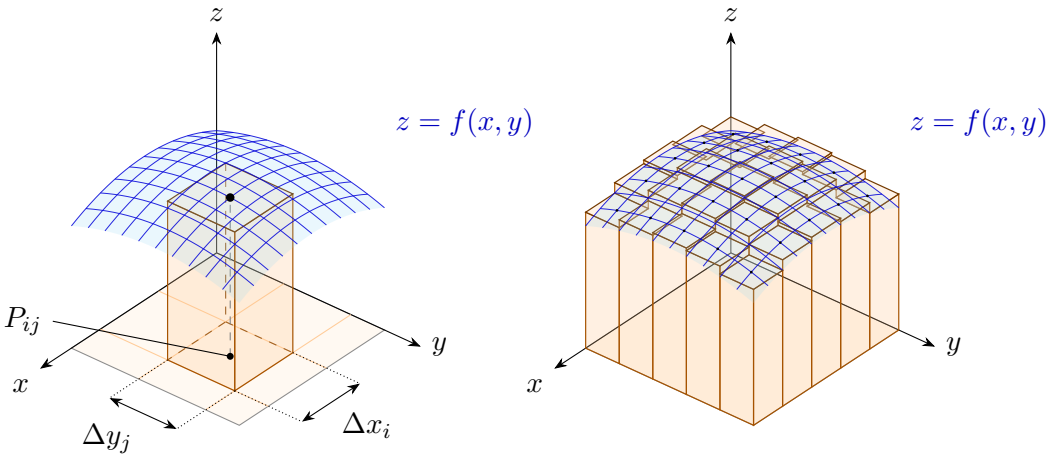
The Riemann sum for a function $f(x)$ over an interval $[a, b]$ is given by

$$\sum_{i=1}^{n} f(p_i)\Delta x_i$$

where $\Delta x_i$ is the width of the $i$-th subinterval, and $p_i$ is a sample point in that subinterval, as indicated in the figure below.




We can extend this idea to functions of two variables. Assume that $\mathcal{D} = [a, b] \times [c, d]$ is rectangular region in the $xy$-plane.




By this, we could write a precise definition of the double integral as follows:

**Definition 11.2: Precise Definition of Double Integral**

Let $f(x, y)$ be a function defined on a closed and bounded region $\mathcal{D}$ in the $xy$-plane.

We partition $\mathcal{D}$ into $m$ subintervals in the $x$-direction and $n$ subintervals in the $y$-direction, forming a grid of rectangles.

Let $\Delta x_i$ and $\Delta y_j$ be the widths of the subintervals in the $x$ and $y$ directions, respectively, and let $P_{ij}$ be a sample point in the rectangle formed by the $i$-th subinterval in the $x$-direction and the $j$-th subinterval in the $y$-direction.

The **double integral** of $f(x, y)$ over the region $\mathcal{D}$ is defined as the limit of the Riemann

sum as the maximum width of the subintervals approaches zero:

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \lim_{\|P\|\to 0} \sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta A_{ij}$$

Notice that the $\Delta A_{ij}$ here does not always exist. If it does, we call the function $f(x,y)$ **integrable** over the region $\mathcal{D}$. Also, if a function is continuous over a closed and bounded region, then it is integrable over that region.

## 11.2 Iterated Integrals and Fubini's Theorem

To evaluate double integrals, we can use Fubini's Theorem, which allows us to compute the double integral as an iterated integral.

**Theorem 11.1: Fubini's Theorem**

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \int_a^b \int_c^d f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_c^d \int_a^b f(x,y)\,\mathrm{d}x\,\mathrm{d}y$$